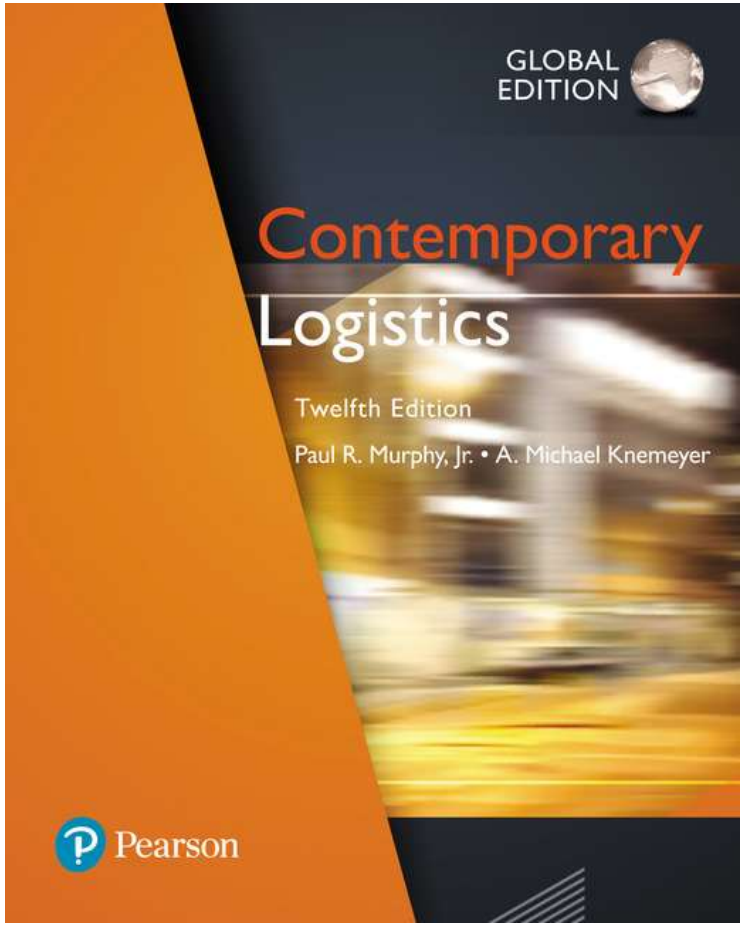
Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 1

An Overview of Logistics

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Learning Objectives

1.1 To discuss the economic impacts of logistics

1.2 To define what logistics is

1.3 To analyze the increased importance of logistics

1.4 To discuss the systems and total cost approaches to logistics

1.5 To expose you to logistical relationships within the firm

1.6 To introduce you to marketing channels

1.7 To provide a brief overview of activities in the logistics channel

1.8 To familiarize you with logistics careers

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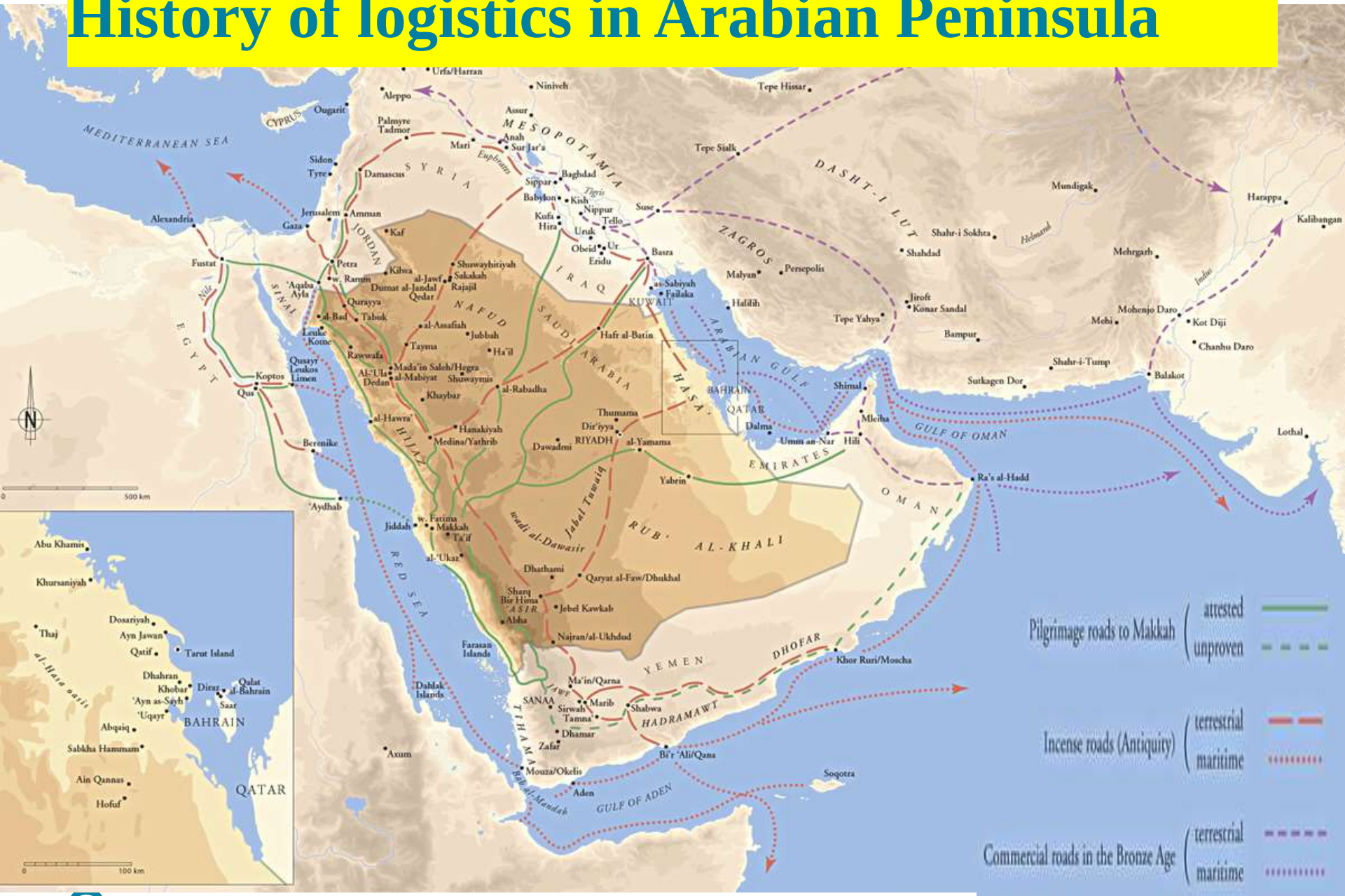
History of logistics in Arabian Peninsula



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Economic Impacts of Logistics

- Macroeconomic impacts
  - Logistics is an important component in any country's economy
- Economic utility: the value, or usefulness, of a product in fulfilling customer needs. It has 4 forms:
  1. Possession utility: value, or usefulness, coming from a customer able to possess a product. **Own or use** a product
  2. Form utility: product being in a form that can be used by the customer and is of value to the customer. A **usable form**
  3. Place utility: products available *where* they are needed by customers . A**vailable at the right location**.
  4. Time utility: having products available *when* they are needed by customers. A**vailable at the right time**.

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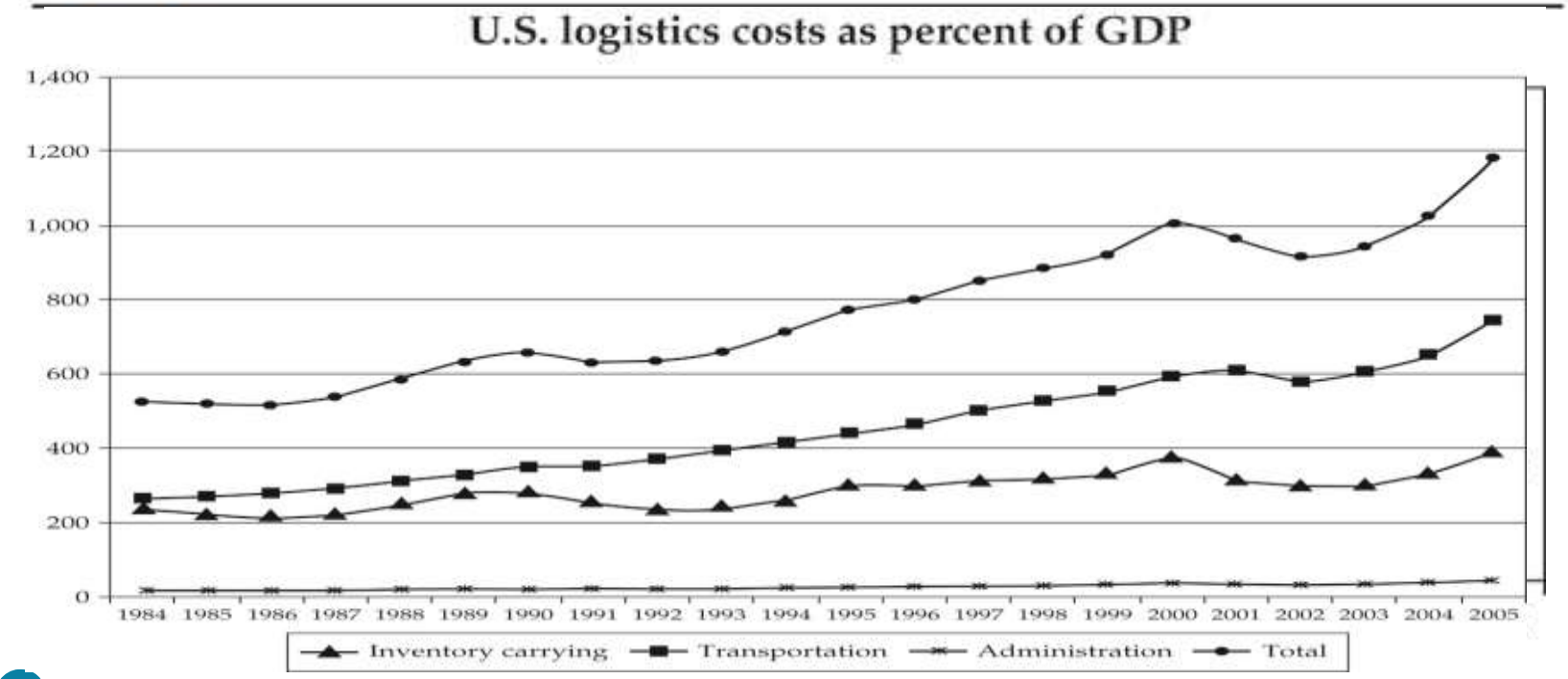
Table 1.1: The Cost of the Business Logistics System in Relation to Gross Domestic Product

Country	Logistics as a Percentage of GDP
United States	8.5
Brazil	12.0
South Africa	12.8
India	13.0
People's Republic of China	18.0
Vietnam	25.0
Indonesia	27.0



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Logistics: What It Is

- Council of SCM Professionals (CSCMP) definition:

"**Logistics management** is part of supply chain management that plans, implements, and controls the efficient, effective forward and reverse flow and storage of goods, services, and related information between the point of origin and the point of consumption in order to meet customers' requirements"[1]

"**Supply chain management** encompasses the planning and management of all activities involved in sourcing and procurement, conversion, and all logistics management activities. Importantly, it also includes coordination and collaboration with channel partners, which can be suppliers, intermediaries, third party service providers, and customers. In essence, supply chain management integrates supply and demand management within and across companies"

[1]www.cscmp.org

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The Increased Importance of Logistics (1 of 2)

- A reduction in economic regulation
  - Allowed individual carriers, e.g., airfreight, railroad, and trucking industries, flexibility in pricing and service. Facilitated the implementation of tailored logistics approach, in the sense that companies could specify different logistics service levels, and prices could be adjusted accordingly
- Changes in consumer behavior
- Technological advances
  - Internet, e-shopping, RFID, shipment tracking GIS, AI, Robotics, Electric vehicles,

Insight: Logistics became important not because companies wanted it to be—but because markets became faster, customers became more demanding, and competition became global."

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The Increased Importance of Logistics (1 of 2)

- Advances in retailing
  - Different channels. e.g., online, in-store, and mobile app. **Omnichannel retailing** is a strategy focusing on providing customers a seamless shopping experience regardless of sales **channel**
- Globalization of trade
  - International logistics
  - Shipping **container** is often championed as an important catalyst for the growth in global trade
  - Geographic distances between buyers and sellers are often greater, monitoring logistics processes is sometimes complicated by differences in business practices, culture, and language

Capacity 24,000 containers

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The Systems and Total Cost Approaches to Logistics (1 of 2)

- Systems approach
  - Compatibility between a company's goals and objectives and major functional area goals and objectives
  - Interdependence of functional areas (marketing, finance, production, and logistics)
    - Example: Stock-keeping units (SKUs)
  - Interdependence of logistics activities or intra-functional logistics between <u>materials management</u> and <u>physical distribution</u>

Almarai's mission to provide quality and nutritious food & beverages that enrich consumers' lives every day

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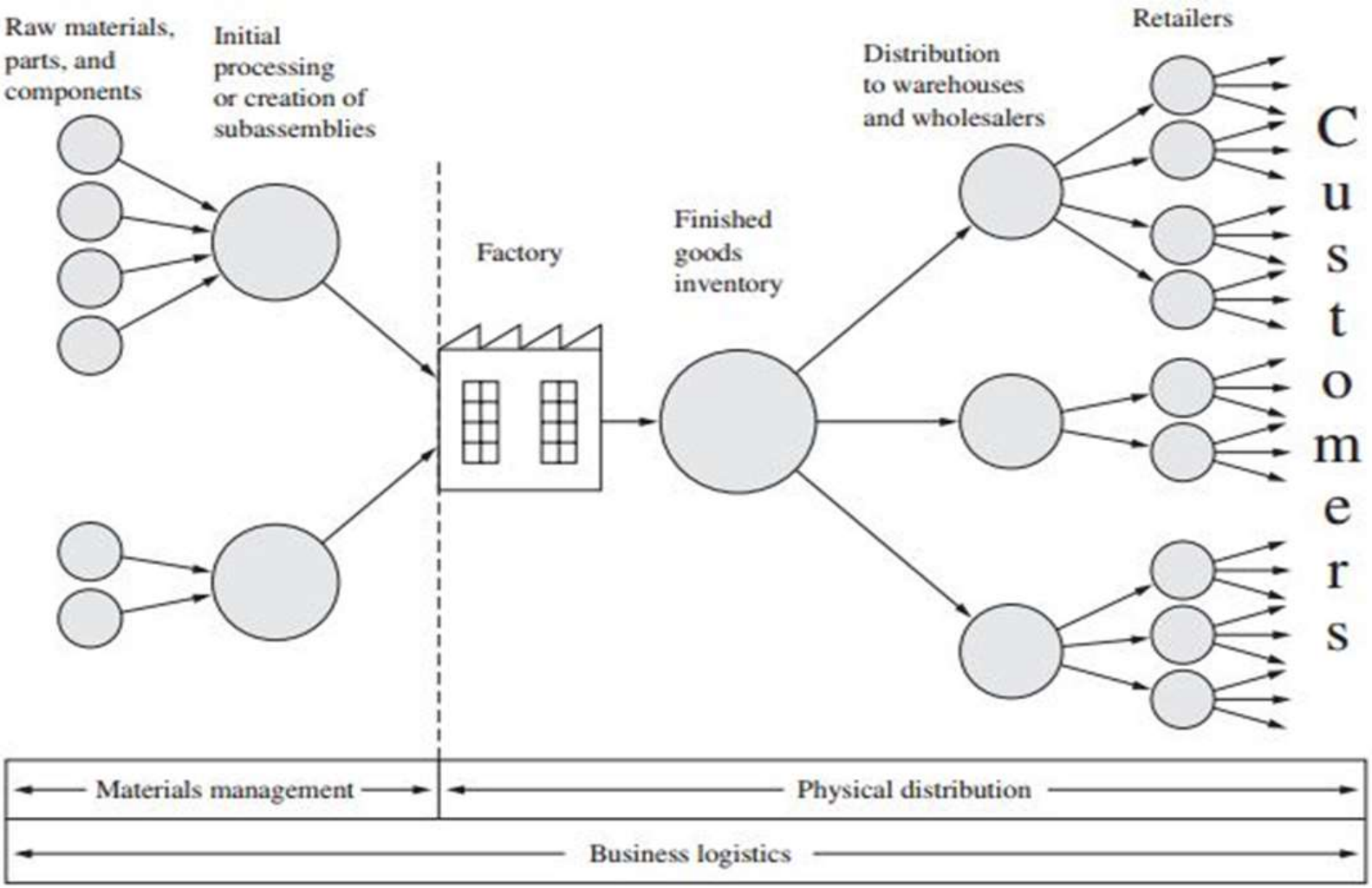
Figure 1.1: Control Over the Flow of Inbound and Outbound Movements



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The Systems and Total Cost Approaches to Logistics (2 of 2)

- Total cost approach
  - Cost trade-offs: changes to one activity cause some costs to increase and others to decrease.
  - Example, a decrease in transportation costs is often associated with an increase in warehousing costs. Why?
  - Total logistics concept: integration of all activities into a unified whole that seeks to minimize distribution costs in a manner that supports an organization's strategic objectives

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Logistical Relationships within the Firm (1 of 2)

- Finance
  - Finance dep. is often instrumental in approving capital budgeting decisions that affect logistics, such as the acquisition of materials handling equipment and packaging equipment
  - Finance department often measures inventory in terms of its cost or value in dollars, whereas logistics tends to measure inventory in terms of units
- Production
  - production favors long production runs of individual products because this allows the fixed costs to be spread over more units, thus resulting in a lower production cost per unit
  - Larger stocks for logistics and some excess stock of slow-selling units
  - **postponement** concept to delay of value-added activities such as assembly, production, and packaging until the latest possible time and possible done in warehouses.

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Logistical Relationships within the Firm (2 of 2)

- Marketing: 4 Ps (Place, Price, Product, Promotion) of decisions
  - Place decisions: place decisions involve new strategies for reaching customers. **Co-branding** is an alliance that allows customers to purchase products from two or more name-brand retailers at one store location.
  - Price decisions: Price influence demand. Landed costs: refers to the price of a product at the source plus transportation costs to its destination. Online merchants require a minimum order amount (e.g., $50) to qualify for “free” delivery.
  - Product decisions:
    - Stockouts dilemma: Marketers prefer to carry high quantities of stocks to avoid stockout. For logistics higher quantities of inventory (1) necessitate additional storage space and (2) increase inventory carrying costs
    - Sustainable products (Fair trades)

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Promotion decisions
Alibaba Singles Day Event 11/11

- 24hour of Sales
- 250,000 brands from 5 millions companies
- 800 million shoppers
- In 2020 $74 billion in sales
- In 2019, 1.3 billion order
- Logistics Network
  - Network of 730 warehouses
  - 3,000 charter flights (within China)
  - 3,000,000 people involved in processing and shipping orders
  - 10,000 lockers delivery (without human contact)
  - 700 robots used in Wuxi warehouse

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Marketing Channels (1 of 2)

- Marketing channels refer to “a set of institutions necessary to transfer the title to goods and to move goods from the point of production to the point of consumption and, as such, which consists of all the institutions and all the marketing activities in the marketing process.”[2]
- Main channel members:
  - Manufacturers
  - Wholesalers
  - Retailers

[2]American Marketing Association Dictionary, www.marketingpower.com

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Marketing Channels (2 of 2)

Channel members work in different arrangements

- Ownership channel
  - Covers movement of the title to the goods
- Negotiation channel
  - Buy and sell agreements are reached
- Financing channel
  - Payments for goods
- Promotions channel
  - Promoting a new or existing product
- Logistics channel
  - Contributes the sorting function to the overall channel process in terms of sorting product, accumulating, allocating, and assorting

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Channel Intermediaries/Facilitators

- Ownership channel
  - Banks, finance companies
- Negotiation channel
  - Brokers
- Financing channel
  - Banks, insurance companies, finance companies
- Promotions channel
  - Advertising agencies, public relations agencies
- Logistics channel
  - Freight forwarders (eg Aramex)

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Activities in the Logistical Channel

- Customer service
- Facility location decisions
- Inventory management
- Order management
- Procurement
- Transportation management
- Demand forecasting
- International logistics
- Materials handling
- Packaging
- Reverse logistics
- Warehousing management

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Logistics Careers (1 of 2)

- Entry-level position examples include:
  - Logistics (supply chain) analyst
  - Consultant
  - Customer service manager
  - Fulfillment supervisor

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Logistics Careers (2 of 2)

- Second-level position examples include:
  - International logistics manager
  - Supply chain software manager
  - Purchasing manager
  - Transportation manager
  - Warehouse operations manager

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Key Terms (1 of 2)

- 3D printing
- Big-box retailer
- Co-branding
- Container
- Cost trade-offs
- Disintermediation
- Economic utility
- Form utility
- Humanitarian logistics
- Landed cost
- Logistics
- Marketing channels
- Mass logistics
- Materials management
- Omnichannel retailing

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Key Terms (2 of 2)

- Physical distribution
- Place utility
- Possession utility
- Postponement
- Sorting function
- Stock-keeping units (SKUs)
- Stockouts
- Sustainable products
- Systems approach
- Tailored logistics
- Time utility
- Total cost approach

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Contemporary Logistics

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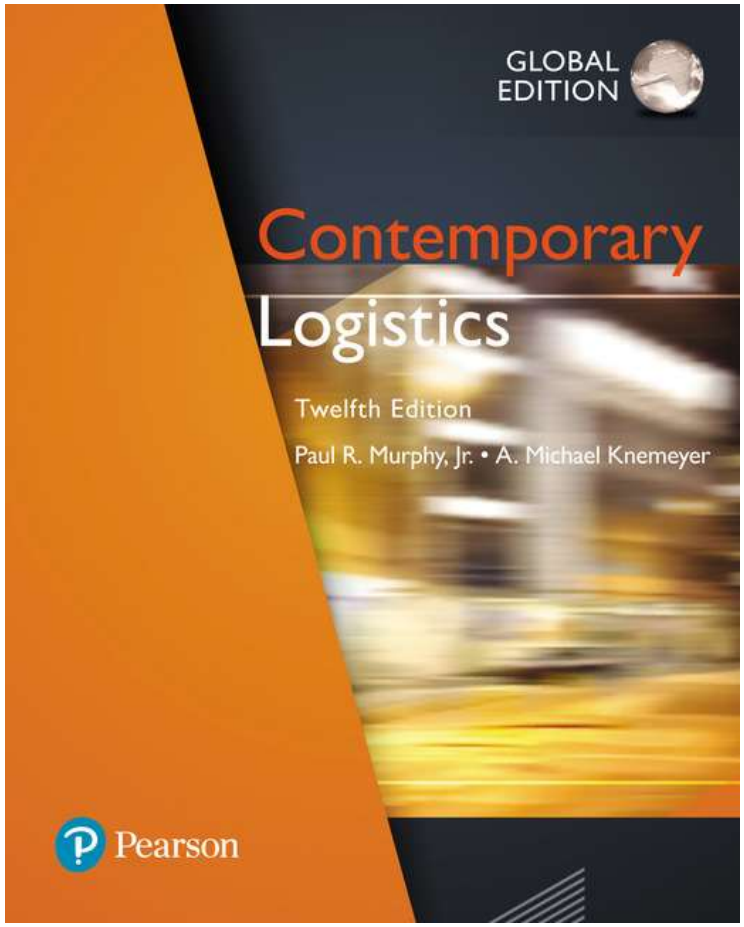

Chapter 4

Organizational and Managerial Issues in Logistics

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Practical Scenario

You are the Head of Logistics in a growing company.

How do you organize people, improve productivity, manage quality, reduce risk, and deal with returns?

Learning Objectives

4.1 To explain organizational structure for logistics

4.2 To compare traditional and modern organizational design for logistics

4.3 To identify productivity issues and improvement efforts in logistics

4.4 To discuss quality issues in logistics

4.5 To describe ways to manage theft and pilferage

4.6 To review the concept of logistics social responsibility

4.7 To articulate issues associated with reverse logistics

4.8 To report on programs designed to lessen the impact of terrorism on logistics systems

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Organizing Logistics Within the Firm (1 of 8)

- Logistics organization: How a company arranges its people, roles, responsibilities, and decision-making to manage the flow of goods and information.
- Logistics organization in a firm depends on number of customers and their locations and the company size
- Two key organizational logistics topics
  - Organizational structure: *what tasks exist and who is responsible for them.*
    - focuses on how work roles and administrative mechanisms are allocated to integrate and control work.
  - Organizational design: *who has power and decision rights?*
    - detail reporting relationships in an organization.
    - who makes work-related decisions and the appropriate communication channels between workers and managers

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Organizing Logistics Within the Firm (2 of 8)

- Organizational structure for logistics focuses on work ‘roles’ are integrated and controlled
- Two basic structures include:
  - Fragmented logistics structure: Logistics activities are managed in multiple departments throughout an organization.

```
Marketing      Operations      Production
    |              |               |
Customer       Transportation    Warehousing
Service
```

  - Unified logistics structure: Multiple logistics activities are combined into and managed as a single department

Fragmented:
- Conflicting goals
- Poor coordination

Unified:
- Shared goals
- Better coordination

```
          Logistics Manager
                |
   -------------------------------
   |            |               |
Transport   Warehousing   Customer Service
```

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Organizing Logistics Within the Firm (3 of 8)

Where logistics decisions are made?

- Organizational structure for logistics. Two basic structures for logistics department include:
  - Centralized logistics organization: Company maintains a single logistics department that administers the related activities for the entire company from the home office.
  - Decentralized logistics organization: Logistics-related decisions are made separately at the divisional or product group level and often in different geographic regions

```
          HQ Logistics
               |
   -------------------------
   |           |           |
Region A    Region B    Region C
```

```
Region A      Region B      Region C
   |             |             |
Logistics     Logistics     Logistics
 Decisions     Decisions     Decisions
```

Efficiency vs Customer responsiveness

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Organizing Logistics Within the Firm (3 of 8)

Discussion:

Market Type	What it Means	Best Logistics Structure
Stable	Predictable demand	Centralized
Volatile	Unpredictable changes	Decentralized
Homogeneous	Similar customers	Centralized
Heterogeneous	Different customers	Decentralized

Would you centralize or decentralize logistics for an international e-commerce company? Why?

international e-commerce faces different customers and fast-changing demand, it needs centralized planning but local execution.
An international e-commerce company should adopt a hybrid logistics structure, centralizing strategic decisions to achieve efficiency while decentralizing operational decisions to ensure responsiveness to local market conditions.

Centralize at HQ: Logistics strategy, technology platforms, Global carrier contracts, Performance standards, inventory policies
Decentralize locally: Last-mile delivery decisions, Customer service responses, Local inventory adjustments, Handling returns

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Organizing Logistics Within the Firm (4 of 8)

- Organizational structure for logistics: Job title or corporate rank (eg manager, director, vice president, chief ranks)
  - Leading-edge organizations tend to head the logistics department by senior-level personnel
  - Generally excluded from holding a “C-level” position

Discussion: Why do you think logistics is usually not a C-level position, despite its strategic importance?

- Three primary types of organizational design include:

Traditional

  - Hierarchical (functional): Top-down flow

Modern

  - Matrix: Cross-functional responsibilities
  - Network: Process philosophy focused on combining tasks into value-creating products and activities

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Organizing Logistics Within the Firm (6 of 8)

- Organizational design for logistics: Network organizational design is manifested in terms of:

Relevancy	**Responsiveness**	**Flexibility**

  - Relevancy: How logistics decisions match current and future customer needs?
    - Refers to satisfying current and emerging customer needs
    - Facilitated by developing mutually beneficial relationships with key customers
    - The relationships should provide an understanding of customer needs and wants

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Organizing Logistics Within the Firm (7 of 8)

- Responsiveness: *doing things quickly when conditions change*
  - Reflects the degree to which an organization can accommodate unique or unplanned customer requests
  - Achieved when appropriate decision makers are provided with both relevant information and the authority to address unique or unplanned requests.

Information + Authority → Responsiveness

Discussion: Why might a highly automated logistics system still be unresponsive?

- No decision authority
- Too many approvals

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Organizing Logistics Within the Firm (8 of 8)

- Flexibility: Is an organization's ability to address unexpected operational situations
  - Predicated on avoiding early commitment to an irreversible course of action
  - Example: the postponement of assembly, labeling, and so on until exact customer requirements are known.

Discussion: Can logistics be efficient but not relevant? Give an example?

Efficient = low cost, fast, well-organized

Relevant = matches what the customer actually wants

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Managerial Issues in Logistics (1 of 28)

Managerial issues in logistics are the ongoing challenges managers face once the structure is in place.

- Companies recognize that logistics systems must not only be organized, but they must also be managed:
  - Productivity
  - Quality
  - Risk
  - Sustainability
  - Complexity

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Managerial Issues in Logistics (2 of 28)

If logistics costs increase but service stays the same, what managerial issue is this?

- Productivity:
  - Can be defined as the amount of output (work done) divided by the amount of input (labors)
  - Provides insight into the efficiency with which corporate resources are being **utilized**

Productivity tells us how well logistics resources are being used, but it does not tell us whether customers are satisfied.

Example: Warehouse A:10 workers1,000 orders/day

Productivity = 100 orders per worker

If the same 10 workers process 1,200 orders, productivity increases — even if cost stays the same.

Question: Can productivity increase while service quality decreases?

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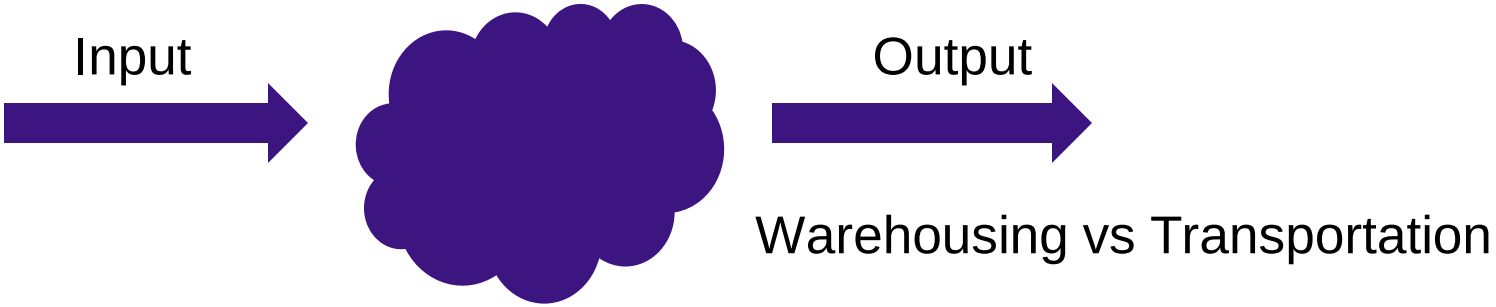
Managerial Issues in Logistics (3 of 28)

- Productivity
  - Three ways to improve productivity include:
    - Reduce the amount of input while holding output constant
    - Increase the amount of output while holding input constant
    - Increase output while decreasing input



Warehouses is about **labor efficiency**, while productivity in transportation is mainly about **asset utilization**.

Discussion : Which productivity improvement approach is most realistic in logistics operations, and why?

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Managerial Issues in Logistics (4 of 28)

- Productivity
  - Labor productivity
    - Warehousing and transportation are heavily dependent on human labor
    - Human labor is an input (i.e., workers receive wages or salaries)
    - **Productivity** improvement efforts in logistics are often directed toward increasing the amount of output while holding input constant (workers resistant to suggestions regarding reduced wages or salaries)

Discussion:

Why do logistics firms prefer increasing output rather than reducing labor to improve productivity?"

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Managerial Issues in Logistics (5 of 28)

- Productivity
  - **In asset productivity**, the concerns include:
    - Space utilization
      - Excess capacity (unused available space)
        - Can be unproductive as it may result in the purchase of additional equipment or facilities, which adds costs (input)
        - May not yield additional output resulting in a productivity decline
    - Improving the output from existing assets
      - Increases productivity as inputs remain constant, but output is increased

Logistics assets include:

- Warehouses
- Trucks
- Trailers
- Material-handling equipment

In warehousing, asset productivity is often measured through **space utilization**

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Managerial Issues in Logistics (7 of 28)

- Quality
  - Logistics service quality
    - Relates to a firm's ability to deliver products, material, and services without defects or errors to both internal (production, sales) and external customers (end customers, retailers)[1]
    - Logistics quality means delivering the right product, in the right condition, at the right time, to the right place, consistently.
    - A logistics system can be productive but still deliver poor quality. Full trucks → late deliveries

Discussion: Which logistics quality failure annoys customers the most: late delivery or wrong delivery? Why?"

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Managerial Issues in Logistics (8 of 28)

- Quality
  - Quality in logistics involves trade-offs
    - If inferior logistic service quality, customers may perceive lower value
    - If superior logistic service quality than expected or required, organization may be adding unnecessary cost
  - Organizations must try to match the quality levels of the logistic services they provide with the expectations of their customers and the landscape in which they operate

Discussion: Logistics quality must match customer expectations — not exceed or fall below them.

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Managerial Issues in Logistics (9 of 28)

- Quality
  - Vendors are expected to have quality programs
  - Vendors can demonstrate commitment to quality to potential buyers through achieving and maintaining quality program certification
    - ISO (International Standards Organization) 9000 certification is an example of a quality program certification

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Managerial Issues in Logistics (10 of 28)

- Quality
  - ISO 9000
    - Is a set of generic standards used to document, implement, and demonstrate quality management and assurance systems
    - Is applicable to both manufacturing and service firms
    - Standards are intended to help companies build quality into every core process in each department
- 9000 does not guarantee perfect quality — it guarantees **consistent processes.**
- To obtain ISO 9000 certification, organizations must document their processes, train employees, and demonstrate continuous improvement

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Managerial Issues in Logistics (11 of 28)

- Quality: ISO 9000
  - Firms demonstrating commitment to quality through **training, reviews, and continuous improvement** achieve initial ISO 9000 certification
  - Once certification is obtained, audits are conducted annually, and organizations can be recertified every three years
  - **Certification is credited with**
    - an increase in customer service
    - improved order accuracy
    - enabling enhanced costs analysis

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Managerial Issues in Logistics (12 of 28)

- Quality
  - Six Sigma
    - Is a quality-focused methodology that emphasizes the virtual **elimination of business errors**
    - Area covered under a normal curve is by six standard deviations is 99.999998 percent
    - Approach suggests that there will be 2 defects, deficiencies, or errors per million units
    - Can be applied to various logistics activities such as order packing

"ISO 9000 focuses on having consistent processes.
Six Sigma focuses on **eliminating errors within those processes.**"

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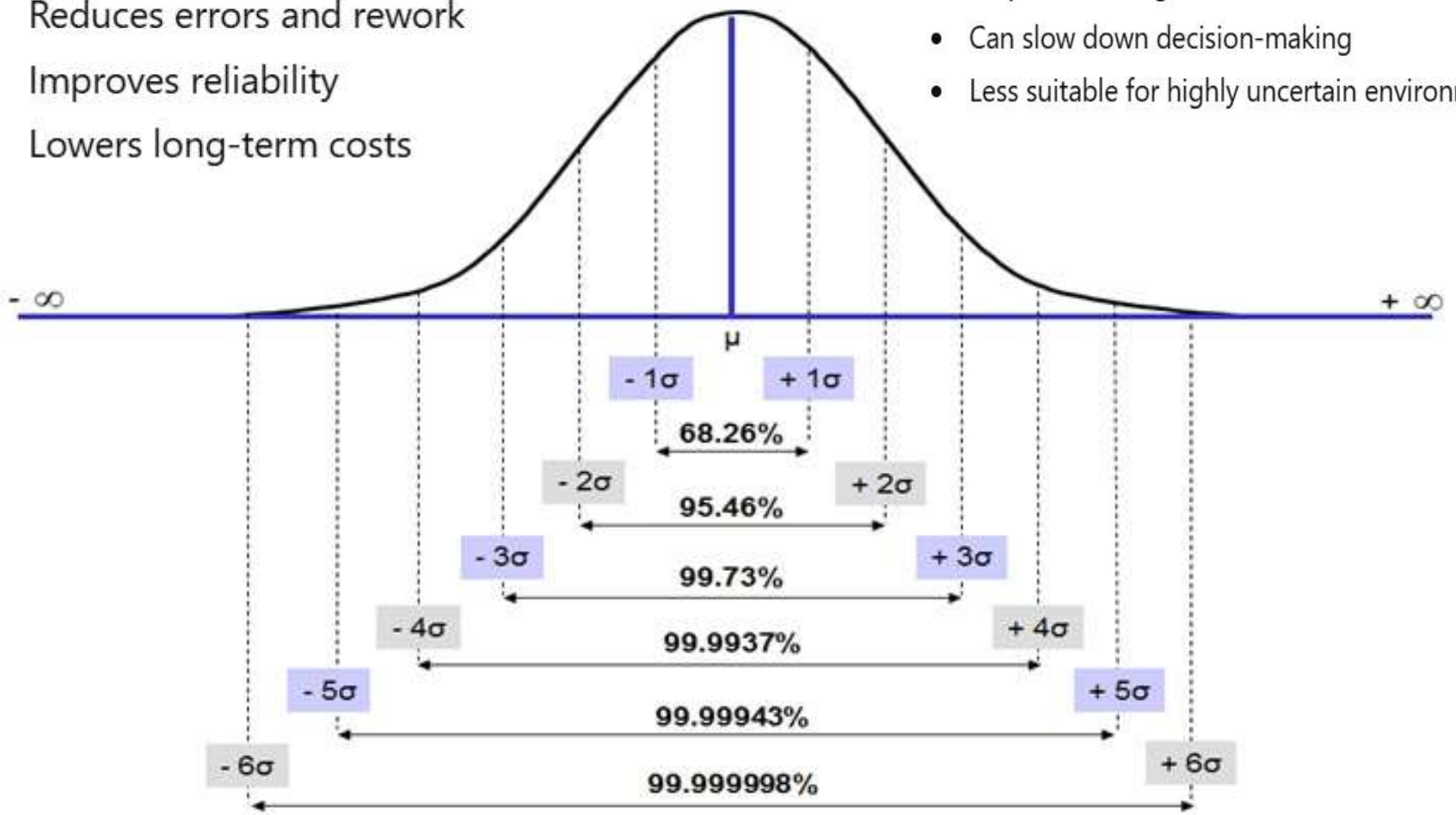
Strengths

- Reduces errors and rework
- Improves reliability
- Lowers long-term costs

Limitations

- Requires training and investment
- Can slow down decision-making
- Less suitable for highly uncertain environments



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Managerial Issues in Logistics (14 of 28)

- Lean → speed & waste reduction
- Six Sigma → accuracy & consistency

```
Lean (Speed) + Six Sigma (Accuracy)
                ↓
        Lean Six Sigma
```

- Quality: Lean Six Sigma
  - Quality-focused approach that integrates **Six Sigma** with the **Lean principle**
  - Integrates goals and methods of these two approaches in pursuit of quality
  - Recognizes that firms cannot focus only on quality or speed
  - Emphasizes an organizational focus on improving quality as it relates to responsiveness

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Managerial Issues in Logistics (13 of 28)

- Quality: Six Sigma
  - Benefits
    - Reduced costs
    - Reduced errors and waste
    - Reduced cycle time
  - Drawbacks
    - Cultural resistance
    - Investing required resources (both human and money)….. Training costs
    - Management commitment

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Managerial Issues in Logistics (16 of 28)

- Risk
  - Disruptions that could lead to interrupt the flow of goods and cause losses for firm.
  - **Types of Logistics Risk**
    - Regularly occurring (or operational) risks (e.g., variability in demand or potential for a damaged shipment (goods))
    - Catastrophic risks (e.g., earthquakes or terrorist attacks)

Discussion :
Is a damaged shipment always a transportation problem?"

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Managerial Issues in Logistics (17 of 28)

- Risk
  - Logistics uncertainty **pyramid model**
    - Established to **identify sources of uncertainty** that can affect the risk exposure for logistics activities
    - Identifies several types of uncertainty including shipper, customer, carrier, control systems, and external

The logistics uncertainty pyramid shows different sources of uncertainty that affect logistics performance, arranged from internal to external.

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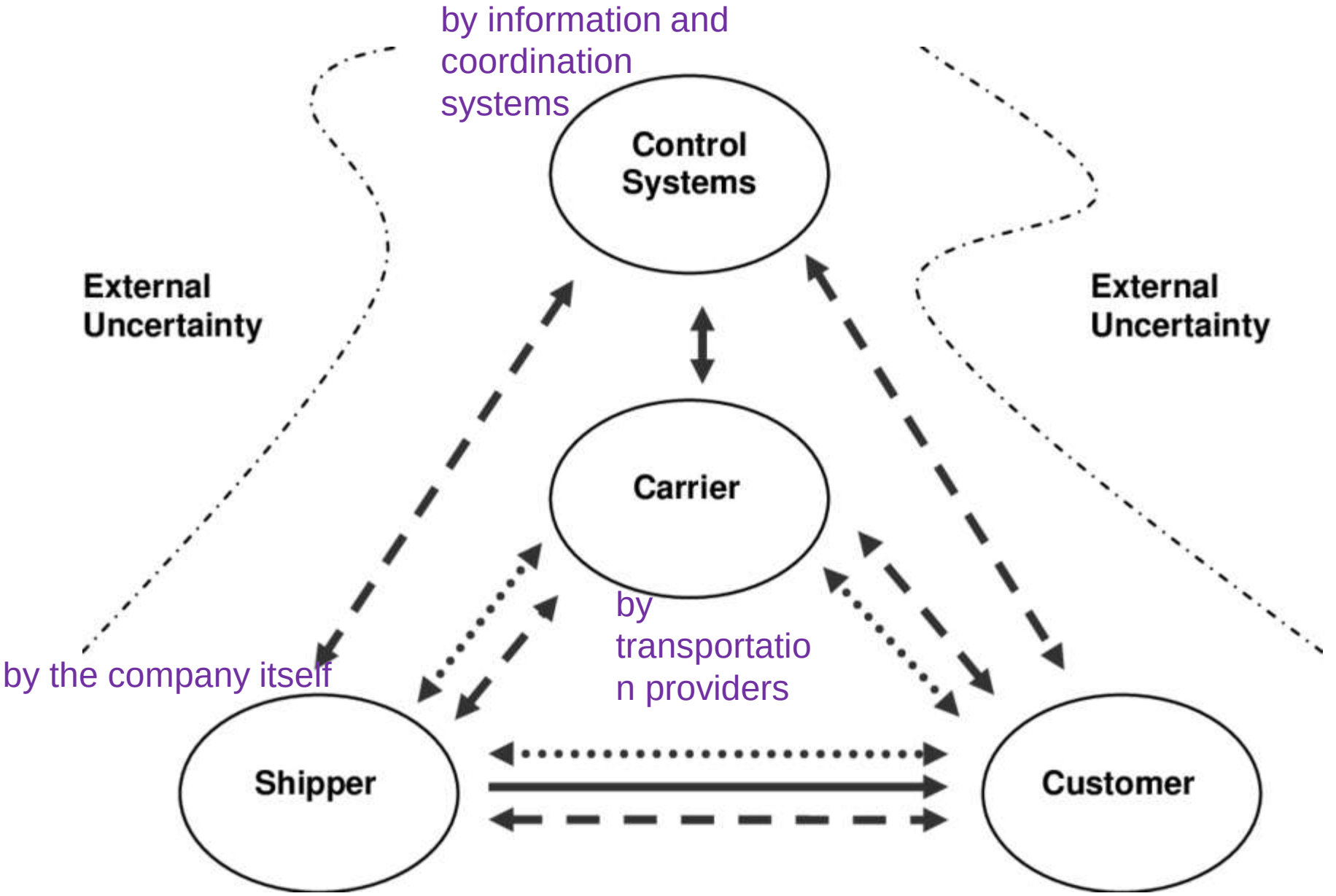

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Managerial Issues in Logistics (18 of 28)

- Risk
  - Terrorism is an example of catastrophic risk
  - Terrorism can be defined as “an illegal use of or threat of force or violence made by a group or individual against a person, a company, or someone’s property with a goal of menacing the target, often grounded in politics or ideology.”[2]

[2]Terrorism, *The American Heritage® Dictionary of the English Language*, 4th ed. (n.d.). Retrieved from Dictionary.com website: *http://dictionary.reference.com/browse/terrorism.*

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Managerial Issues in Logistics (19 of 28)

- Risk
  - Creation of the Department of Homeland Security (DHS)
    - Federal agency
    - Goals
      - To prevent terrorist attacks in the U.S.
      - To reduce the vulnerability of the U.S. to terrorism

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Managerial Issues in Logistics (20 of 28)

- Risk
  - The 22 separate government entities incorporated into DHS include:
    - Transportation Security Administration (TSA)
      - Transportation Worker Identification Credential (TWIC)
    - Customs and Border Protection (CBP)
      - Container Security Initiative (CSI)
      - Customs Trade Partnership Against Terrorism (C-TPAT)
      - Importer Security Filing (ISF) rule, also known as “10+2”

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Table 4.1: Timeline for Presenting Electronic Advance Manifest Information

Inbound to the United States	
Mode	**Timeline**
Air and courier	Four hours prior to arrival in the United States, or "wheels up" from certain nearby airports
Rail	Two hours prior to arrival at a U.S. port of entry
Ocean vessel	24 hours prior to lading at foreign port
Truck	Free and Secure Trade (FAST): 30 minutes prior to arrival in the United States; non-FAST: one hour prior to arrival in the United States
Outbound from the United States	
Mode	**Timeline**
Air and courier	Two hours prior to scheduled departure from the United States
Rail	Two hours prior to the arrival of the train at the border
Ocean vessel	24 hours prior to departure from U.S. port where cargo is laden
Truck	One hour prior to the arrival of the truck at the border

Source: Erlinda Byrd, "Rules for Improving Cargo Security," *Customs and Border Protection Today,* March 2004.

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Managerial Issues in Logistics (21 of 28)

- Risk
  - Theft is another logistics risk issue that confronts many managers
  - Theft (stealing) can be defined as the taking and removing of personal property with the intent to deprive the rightful owner of it.[3]

[3]www.m-w.com/dictionary

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Managerial Issues in Logistics (22 of 28)

- Risk
  - Thoughts regarding theft
    - Insurance companies may reimburse for loss, but time and costs tend not to be covered
    - Theft results in the planned flow of goods being interrupted, which can lead to stockouts
    - Theft can factor into facility location decisions

Freight train looting

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Managerial Issues in Logistics (23 of 28)

- Risk
  - Thoughts regarding pilferage (employee theft)
    - Transportation and warehousing operations are particularly vulnerable to pilferage
    - Managing pilferage begins with the hiring process
    - Zero-tolerance pilferage policy
    - Keep goods moving through the system
    - Recent increase in pirate attacks

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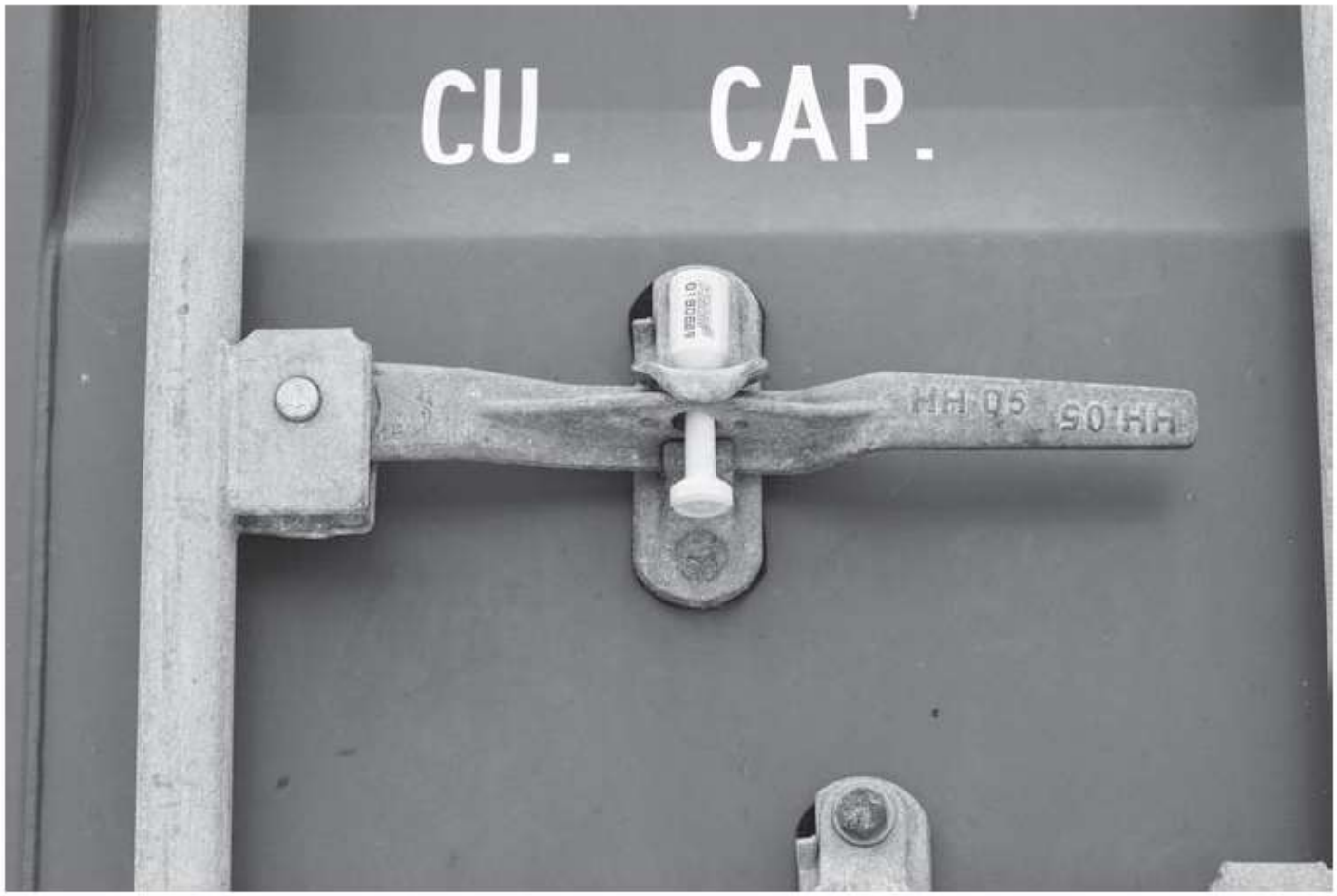
Figure 4.1: Shipping Container Locking Handle with a Uniquely Numbered Customs Seal



Source: Philip Cridland/Alamy Stock Photo

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Managerial Issues in Logistics (24 of 28)

- Sustainability
  - Logistics social responsibility
    - Corporate social responsibility issues that relate directly to logistics

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Managerial Issues in Logistics (25 of 28)

- Sustainability
  - Logistics social responsibility
    - Potential dimensions include:
      - Environment
      - Ethics
      - Diversity
      - Safety
      - Philanthropy
      - Human rights
      - Others

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Managerial Issues in Logistics (26 of 28)

- Sustainability
  - Reverse logistics
    - Is the process of managing return goods
    - Exceeds $100 billion in U.S. alone
    - Can be four to five times more expensive than forward logistics
    - Process can take 12 times as many steps as forward logistics

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Managerial Issues in Logistics (27 of 28)

- Sustainability
  - Reverse logistics
    - Reverse logistics process focuses on:
      - Why products are returned
      - How to optimize reverse logistics
      - Whether reverse logistics should be managed internally or outsourced to a third party[4]

[4]John Paul Quinn, "Are There Ever Any Happy Returns?" *Logistics Management*, June 2005, 63–66.

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Managerial Issues in Logistics (28 of 28)

- Complexity
  - Network complexity
    - Is the growing number of nodes and the associated changes to the links in logistics systems
  - Process complexity
    - Centers on the haphazard development of processes, additions and modifications to processes over time, and/or changing process requirements
  - Range complexity
    - Centers on the implications associated with the increasing number of products that most companies continue to face in an effort to differentiate themselves with their customers

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Key Terms (1 of 3)

- “C-level” position
- Centralized logistics organization
- Container Security Initiative (CSI)
- Customs Trade Partnership Against Terrorism (C-TPAT)
- Decentralized logistics organization
- Excess capacity
- Fragmented logistics structure
- Importer Security Filing (ISF) rule

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Key Terms (2 of 3)

- ISO 9000
- Lean Six Sigma
- Logistics service quality
- Logistics social responsibility
- Logistics uncertainty pyramid model
- Malcolm Baldrige National Quality Award
- Pilferage
- Productivity
- Reverse logistics
- Sharing economy

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Key Terms (3 of 3)

- Six Sigma
- Tachograph
- Theft
- Transportation worker identification credential (TWIC)
- Unified logistics structure

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Activity 1

Task 1

Select ONE peer-reviewed journal article that:

- Is **published in 2020 or later**
- Is **directly related** to the logistics topics studied this week (e.g. logistics organization, productivity, quality, risk, security, sustainability, reverse logistics)

- **Summarize the article** by clearly explaining (no more 2 pages):
  - The research problem or objective
  - The context (industry, country, or supply chain setting)
  - The key findings or arguments

Task2 (no more 2 pages**)**

Identify and critically review FIVE (5) discussions covered in this week's lectures, explaining their relevance to logistics management.

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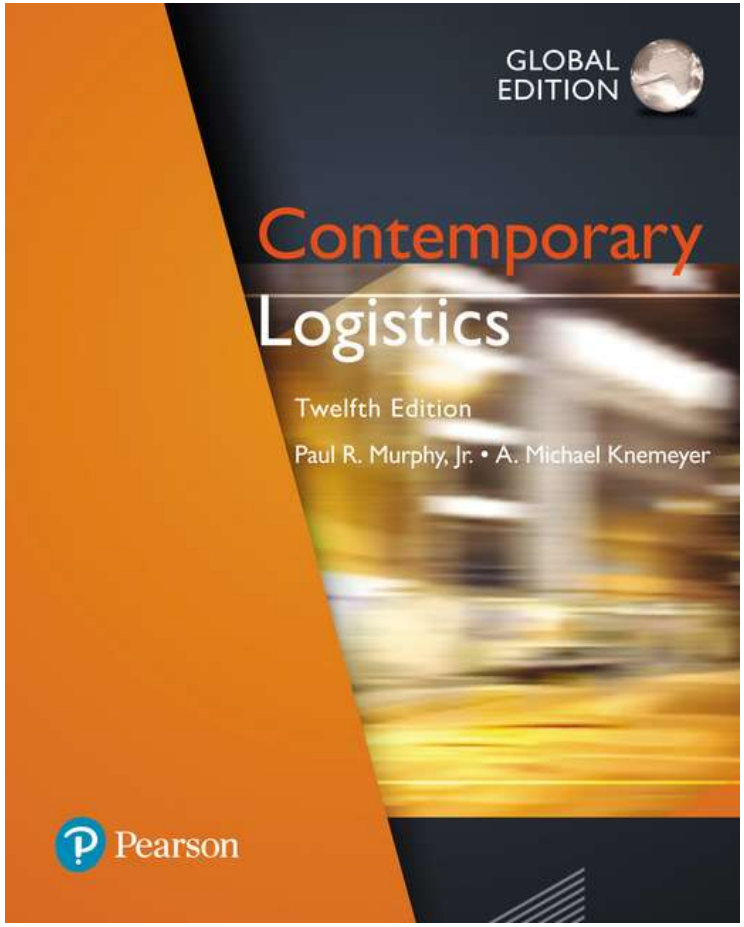

Chapter 6

Procurement

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- This chapter is not theory — this is where money, power, and risk live in the supply chain.

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Learning Objectives (1 of 2)

6.1 To compare procurement, purchasing, and supply management

6.2 To review procurement objectives

6.3 To review supplier selection and evaluation approaches

6.4 To review Kraljic's Portfolio Matrix

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Learning Objectives (2 of 2)

6.5 To establish issues associated with global procurement

6.6 To identify social and environmental aspects of sustainable procurement

6.7 To describe the concept of supply chain finance and how it can be used in procurement relationships

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Procurement (1 of 7)

- When we say *procurement*, we are talking about **everything the company buys from outside** to operate and produce.
- Refers to the raw materials, component parts, and supplies bought from outside organizations to support a company's operations

•Raw materials	• Steel, oil, wheat, chemicals
•Component parts	• Chips, engines, batteries
•Supplies	• Packaging, office supplies, MRO items

Discussion: If procurement controls 70% of cost, **why do most CEOs come from marketing or finance, not procurement?**"

- Procurement costs often range between 60 and 80 percent of an organization's revenues

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Procurement (2 of 7)

- Currently has a more **strategic orientation** in many organizations
- Procurement manager may have responsibility for:
  - Reducing cycle times
  - Playing an integral role in product development
  - Generating additional revenues through collaboration with the marketing department[1]

Procurement is no longer about buying cheap — it is about **competing better.**

Discuss: Which of these three roles aboveis the *hardest* for procurement to achieve in real companies, why?

[1]John Hyatt, "The Rise and Rise of Procurement," *CFO* 28, no. 4 (2012): 57–59.

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Procurement (3 of 7)

- Is procurement the same as purchasing — or something bigger?
- "Procurement" and "purchasing" are sometimes viewed as synonymous terms.

Purchasing is a *subset* of procurement.

- **Purchasing:** Transaction-focused, Short-term, Operational.
- **Procurement:** Strategic + operational, Medium to long term, Cross-functional
- Supply management is viewed as a relational exchange approach involving a limited number of suppliers

Purchasing → Procurement → Supply Management

(Transaction) (Strategy) (Relationships)

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Procurement (4 of 7)

Digital tools don't just make procurement faster — they change **behavior, transparency, and negotiation dynamics**

- Electronic commerce continues to bring many changes to the procurement discipline including
  - Electronic procurement
    - Procurement cards (p-cards)
      - ➢ Push control after purchase (monitoring)Instead of before purchase (approvals)
  - Reverse auctions:
  - Reverse auction = **buyers bid, suppliers compete**.
  - Instead of :Suppliers quote prices → buyer chooses

•Electronic procurement:

✓ Buying through **digital platforms**

✓ Automated workflows

✓ System-based approvals

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Procurement (5 of 7)

- Procurement cards (p-cards)
  - Similar to Visa and MasterCard, but used for an organization's buying needs
  - Number of authorized employees is limited
  - Detailed statement is provided each month
  - Can implement control processes that:
    - Measure usage
    - Identify procurement trends
    - Limit spending during a procurement cycle
    - Block unauthorized expenditures

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Procurement (6 of 7)

- Benefits of procurement cards (p-cards)
  - Reduction in the number of invoices
  - Firm will make one payment each month regardless of the number of p-card holders
  - Allows employees to make purchases in minutes versus days
  - Generally allow suppliers to be paid in a more timely fashion

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Procurement (7 of 7)

- Challenges of procurement cards (expansion overseas)
  - Currency differences
  - Availability of technology
  - Difference in card acceptance
  - Cultural issues with the program

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Procurement Objectives

- Supporting organizational goals and objectives
- Managing the purchasing process effectively and efficiently
  - Effectiveness = buying the **right thing**
    Efficiency = buying it with **minimum waste**
- Managing the supply base:
  - More suppliers ≠ less risk
- Developing strong relationships with other functional groups
- Supporting operational requirements

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Supplier Selection and Evaluation (1 of 3)

- One of procurement's most important responsibilities
- Involves stating an organization's needs and determining how well various potential suppliers can fulfill these needs

Procurement is not choosing a price — it is choosing a *partner* for risk and performance

- **Discussion:** Which is more dangerous: choosing a cheap supplier who fails, or an expensive supplier who performs reliably?

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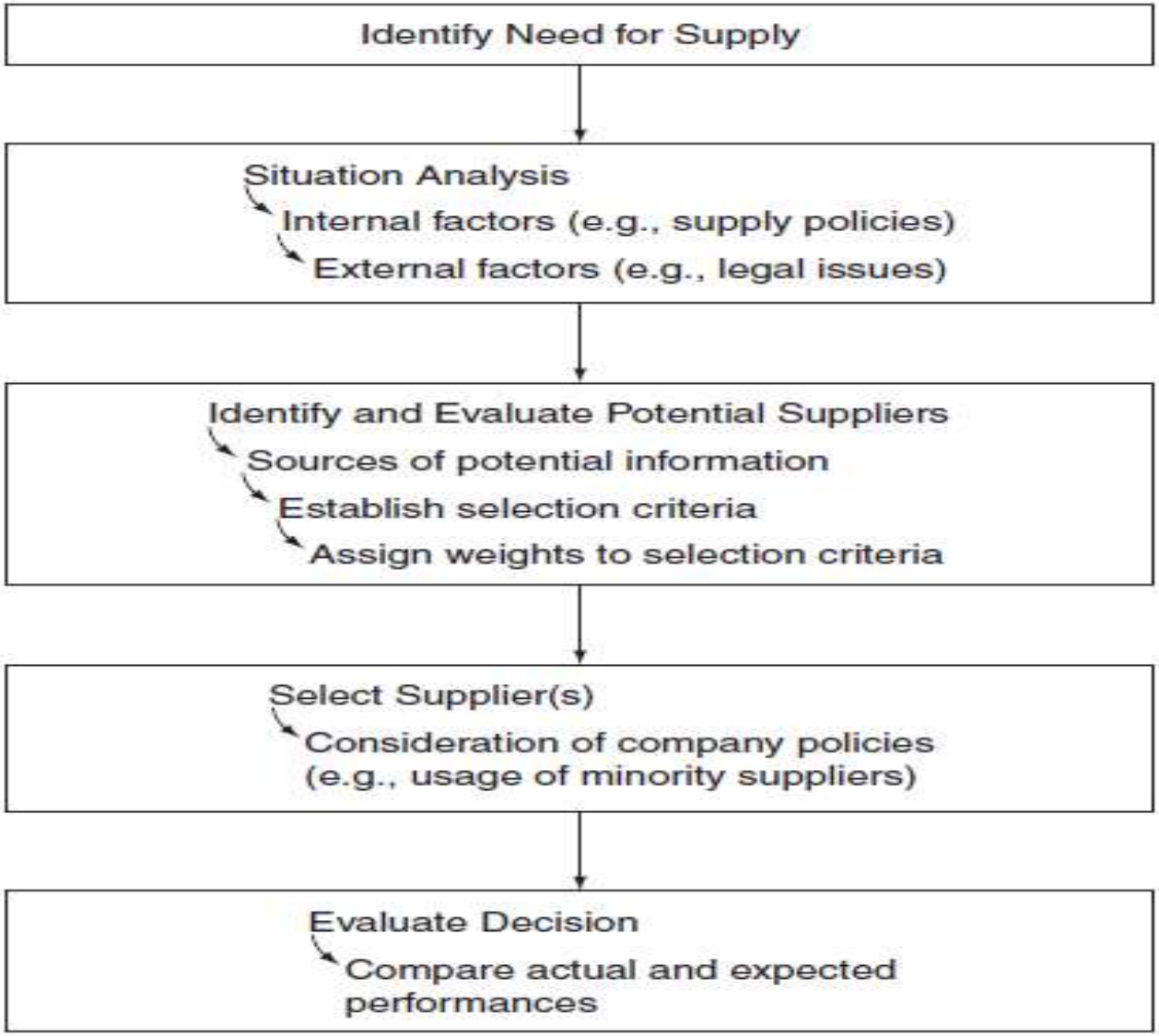
Figure 6.1: Supplier Selection Framework



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Supplier Selection and Evaluation (2 of 3)

- Steps to supplier selection and evaluation
  1. Identify need for supply
  2. Perform situation analysis
  3. Identify and evaluate possible suppliers
  4. Select suppliers
     - Single-source approach
     - Multiple-source approach
  5. Evaluate decision
     - Process-based approach
     - Performance-based approach

Discuss: Which step do you think companies skip most often — and what are the consequences?

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Supplier Selection and Evaluation (3 of 3)

- Evaluate decision
  - Process-based approach
    - Is an assessment of the supplier's service and/or production process (supplier audit)
  - Performance-based approach
    - Is focused on the supplier's actual performance on a variety of criteria including cost and quality
    - Many companies use supplier scorecards to report performance information

Discuss: Would you trust a supplier with perfect processes but poor performance — or great performance but weak processes?

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Procurement Portfolio Approach (1 of 4)

- Within the “situation analysis” step for supplier selection and evaluation, procurement managers must be continually aware of the supply and demand characteristics of the:
  - Raw materials
  - Component parts
  - Purchased supplies

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Procurement Portfolio Approach (2 of 4)

- Kraljic's Portfolio Matrix[2]
  - Used to classify corporate purchases in terms of their importance and supply complexity
  - Goal is to minimize supply vulnerability and getting the most out of the firm's purchasing power

[2]Peter Kraljic, "Purchasing Must Become Supply Management," *Harvard Business Review* 61, no. 5 (1983): 109–117.

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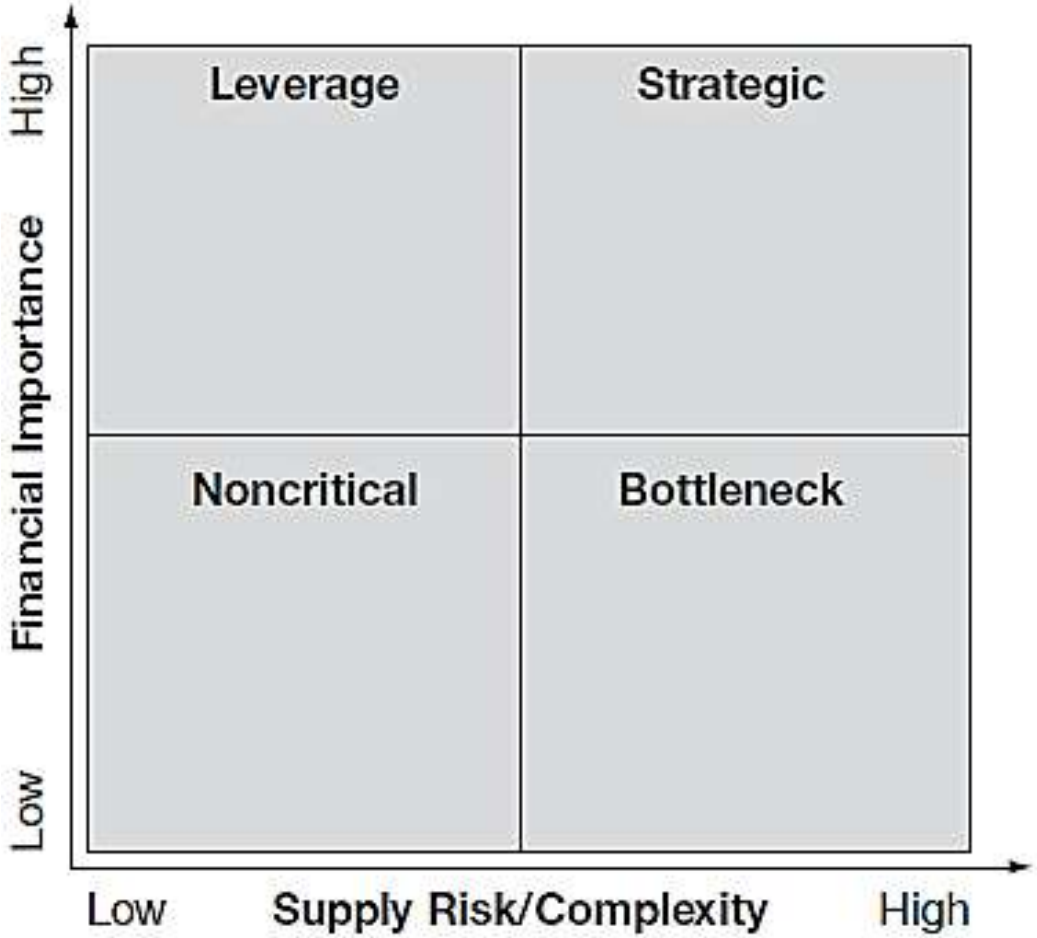
Figure 6.2: Kralijic Portfolio Matrix



Source: Peter Kraljic, “Purchasing Must Become Supply Management.” *Harvard Business Review* 61, no. 5 (1983): 109–117.

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Procurement Portfolio Approach (3 of 4)

- Supplier development (reverse marketing)
  - Refers to aggressive procurement involvement not typically part of supplier selection
  - Can include:
    - Purchaser initiating contact with supplier
    - Purchaser establishing prices, terms and conditions, and other behaviors

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Procurement Portfolio Approach (4 of 4)

- Supplier development (reverse marketing)
  - Motivation to adopt supplier development includes:
    - Numerous inefficiencies associated with suppliers initiating marketing efforts toward purchasers
    - Purchaser may be aware of important benefits which are unknown to the supplier
    - Compel suppliers to meet necessary requirements to achieve competitive advantage in the supply chain

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Break: How Artificial Intelligence Revolutionizes the Procurement Industry

- Automated Supplier Selection
- Demand Forecasting
- Spend Analytics
- Supplier Risk Assessment
- Contract Management
- Fraud Detection
- Dynamic Pricing Optimization
- Automated Purchase Order Processing
- AI-Driven Negotiation Support
- Inventory Optimization
- Supplier Relationship Management
- Sustainable Procurement

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Global Procurement (Sourcing) (1 of 3)

- Refers to buying components and inputs anywhere in the world
- Driven by:
  - Factor-input strategy (organization is seeking low-cost or high-quality sources of supply)
  - Market access strategy (organization is sourcing in markets where it plans to do significant business)

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Global Procurement (Sourcing) (2 of 3)

- Components of global sourcing development model:
  - Planning
  - Specification
  - Evaluation
  - Relationship management
  - Transportation and holding costs
  - Implementation
  - Monitoring and improving

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Global Procurement (Sourcing) (3 of 3)

- Challenges in establishing a successful global sourcing strategy include understanding hidden costs as supply bases are expanded
- Examples of hidden costs:
  - Increased costs of dealing with suppliers outside the domestic market
  - Duty and tariff changes that occur over supply agreement life
  - Increased inventory-related costs associated with global supply chains
  - Rising levels of logistics cost volatility (e.g., ocean freight rates)

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Sustainable Procurement (1 of 7)

- Refers to the integration of social and environmental considerations into all stages of the purchasing process
- Goal is to minimize the impact of procurement activities on human health and the environment

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Sustainable Procurement (2 of 7)

- Social responsibility
  - Socially responsible procurement consists of five dimensions:
    - Diversity
    - The environment
    - Human rights
    - Philanthropy
    - Safety

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Sustainable Procurement (3 of 7)

- Social responsibility
  - Ethical considerations
    - “Win-at-all-costs” philosophy can exacerbate unethical behavior

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Sustainable Procurement (4 of 7)

- Social responsibility
  - Areas of ethical concern in procurement:
    - Gift giving and receiving
    - Bribes and kickbacks
    - Misuse of information
    - Improper methods of knowledge acquisition
    - Lying or misrepresentation of the truth
    - Product quality (lack of)
    - Misuse of company assets
    - Conflicts of interest

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Sustainable Procurement (5 of 7)

- Investment recovery
  - Identifies opportunities to recover revenues or reduce costs associated with:
    - Scrap materials
    - Surplus materials
    - Obsolete materials
    - Waste materials
  - Often the responsibility of the procurement manager

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Sustainable Procurement (6 of 7)

- Investment recovery
  - Can provide an organization the opportunity to simultaneously do well and do good in the sense that investment recovery:
    - Increases a seller's revenues (reduces a seller's cost)
    - While addressing selected environmental considerations

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Sustainable Procurement (7 of 7)

- Investment recovery
  - Even the best-managed organizations generate:
    - Excess (surplus)
      - Stock materials that exceed reasonable requirements
    - Obsolete
      - Materials not likely to ever be used again
    - Scrap
      - Materials that are no longer serviceable, have been discarded, or are a by-product of the production process
    - Waste
      - Materials that have been spoiled, broken, or otherwise rendered unfit for further use or reclamation (have no economic value)

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Supply Chain Finance (1 of 2)

- Refers to a set of technology and financed-based processes that strives to optimize cash flow by:
  - Allowing businesses to extend their payment terms to their suppliers
  - While simultaneously allowing suppliers to get paid early

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Supply Chain Finance (2 of 2)

- Transactions between buyers and sellers are facilitated by financial technology firms (FinTech)
- FinTech companies use cloud-based software to optimize the connection between procurement and accounts payable

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Key Terms (1 of 2)

- Bribes
- Excess (surplus) materials
- FinTech
- Global procurement (sourcing)
- Investment recovery
- Kickbacks
- Kraljic's Portfolio Matrix
- Multiple sourcing
- Near-sourcing
- Obsolete materials
- Procurement
- Procurement cards (p-cards)
- Purchasing

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Key Terms (2 of 2)

- Scrap materials
- Single sourcing
- Strategic sourcing
- Sustainable procurement
- Supplier audit
- Supplier development (reverse marketing)
- Supplier scorecards
- Supply chain finance
- Supply chain management
- Total cost of ownership (TCO)
- Waste materials

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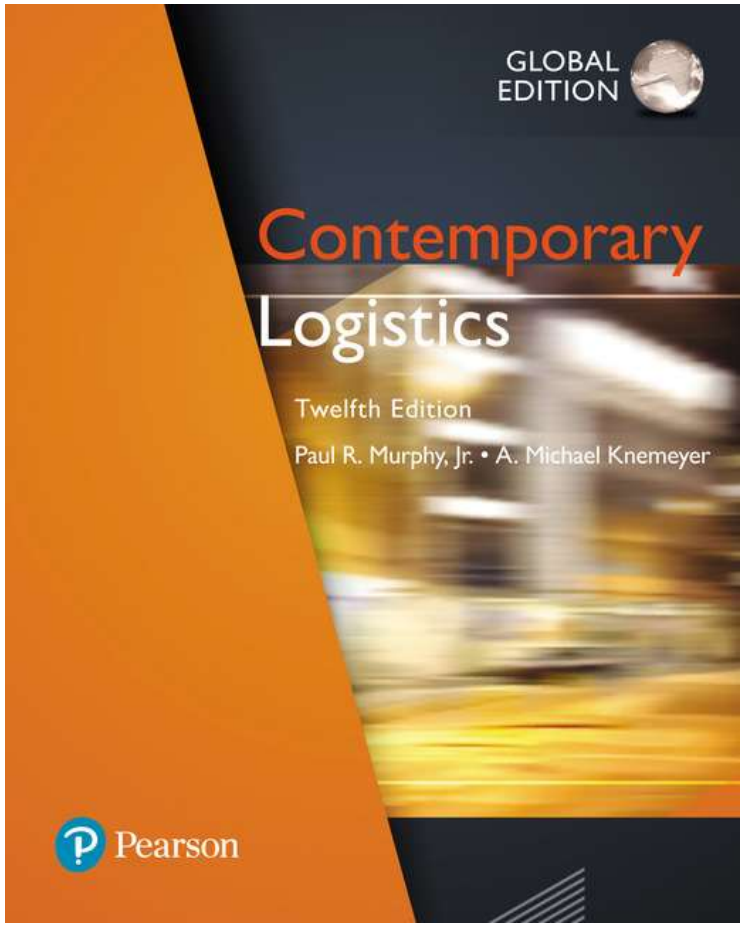

Chapter 7

Demand Management, Order Management, and Customer Service

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Learning Objectives

7.1 To understand the linkages between demand management, order management, and customer service

7.2 To introduce you to demand forecasting models (SKIP)

7.3 To examine the order cycle and its four components

7.4 To understand the four dimensions of customer service as they related to logistics

7.5 To familiarize you with select managerial issues associated with customer service

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Demand Management, Order Management, and Customers Service

(1 of 7)

- Demand management determines how an organization determines what the customer wants. Can be defined as "the creation across the supply chain and its markets of a coordinated flow of demand."[1] =Deciding what to prepare?
- Order management refers to management of the various activities associated with the customers order from start to end. = executing the promise
- Customer service strives to keep customers happy and creates in the customer's mind the perception of an organization with which it is easy to do business. = *how the customer remembers you*

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Demand Management (2 of 7)

- Key component of demand management is **demand (sales) forecasting**
  - Refers to an effort to project future demand
  - Is helpful in **make-to-*stock*** situations and **make-to-*order*** situations
- Three basic types of demand forecasting models:
  - Judgmental: expert opinion
  - Time series: historical data
  - Cause and effect (associative): promotions, price

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Example : Demand uncertainty and aggregation)

2.8 Table 2.60 reports an estimate of the annual mean demand and the corresponding variance of fruit and vegetable products of Flanders, Belgium. The products are divided into five categories: fall–winter, spring–summer, exotic fruits, citrus fruits and dried fruits. Verify whether the aggregate forecast is more or less accurate than the estimated demand for each category

Table 2.60: Annual mean demand and variance of fruit and vegetable products of Flanders

Category	Mean (μ)	Variance (σ^2)
Fall–winter	123,000	24,570
Spring–summer	245,000	35,650
Exotic fruits	9,860	320
Citrus fruits	98,000	2,980
Dried fruits	2,450	456

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Example

Step 1: Aggregate demand statistics

- Mean demand (aggregate):

μ_{agg}=123,000+245,000+9,860+98,000+2,450=478,310 t

- Variance of aggregate demand

σ^2_{agg}=24,570+35,650+320+2,980+456=63,976

$\sigma_{agg}\approx$253

Aggregation reduces relative uncertainty.

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Example

Step 2: Coefficient of variation for each category (CV)

Category	Mean (μ)	Variance (σ^2)	σ	CV (σ/μ)
Fall–winter	123,000	24,570	157	157/123000 ≈0.00128
Spring–summer	245,000	35,650	189	189/245,000 ≈0.00077
Exotic fruits	9,860	320	17.9	17.9/9860 ≈0.00182
Citrus fruits	98,000	2,980	54.6	54.6/98000≈0.00056
Dried fruits	2,450	456	21.4	21.4/2450≈0.00873

Step 3: Coefficient of variation of aggregate demand

$$CV_{agg}=253/\ 478{,}310\approx 0.00053$$

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Example

Step 4: Comparison and conclusion

- The **aggregate coefficient of variation (0.00053)** is **lower than the CV of every individual category**
- Therefore, the **aggregate forecast is more accurate in relative terms**

Conclusion

The aggregate forecast is **more accurate** than the forecast for any individual product category.

Aggregation reduces relative demand variability because fluctuations across categories partially offset each other.

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Order Management (1 of 10)

From click to delivery — where does time get wasted?

- Order cycle (replenishment cycle or lead time) refers to the time from when a customer places an order to when goods are received
- Order management refers to management of the various activities associated with the order cycle
- Order to cash cycle refers to the length of time it takes an organization to receive payment for an order
  - Some organizations include order to cash cycle in their order management model

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Order Management (2 of 10)

- Order cycle time should be analyzed not only in terms of total cycle time but in cycle time variability (reliability) as well
- Order management has been profoundly affected by advances in information systems and tracking and tracking and tracing (HungerStation Order)
- Four stages of the order cycle include:
  1. Order transmittal
  2. Order processing
  3. Order picking and assembly
  4. Order delivery

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Order Management (3 of 10)

1. Order transmittal Place order…..seller receive order
   - Refers to the time from when the customer places an order until the seller receives the order
   - Methods of order transmittal
     - In person (slow)
     - Mail, Telephone, Fax (subject to errors)
     - Electronically online and via mobile apps (accurate and fast but data privacy issue)
2. Order processing Seller receive order …..customer receive order
   - Refers to the time from when the seller receives an order until an appropriate location (i.e., warehouse) is authorized to fill the order

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Order Management (4 of 10)

2. Order processing

- Order processing activities include:
  - Checking for completeness and accuracy
  - Checking the buyer's ability to purchase
  - Order entry into the computer system
  - Crediting salesperson with the sale
  - Recording the transaction
  - Determining inventory location
  - Arranging for outbound transportation
- Some of these activities are sequential (should be done in order) others can be done parallel. This helps in reduce order cycle time

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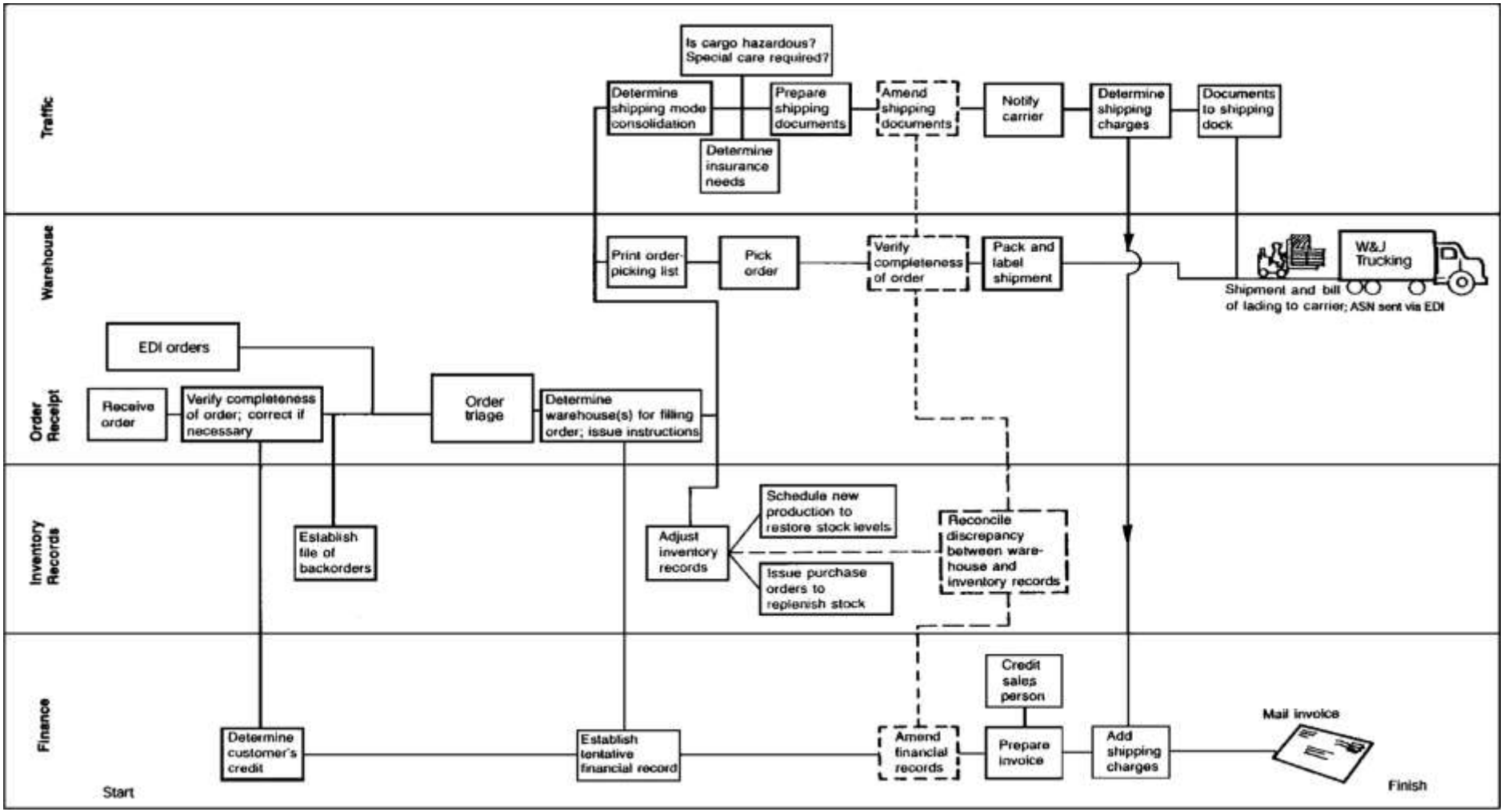
Figure 7.1: Flowchart of Order Handling (2. Order Processing) System



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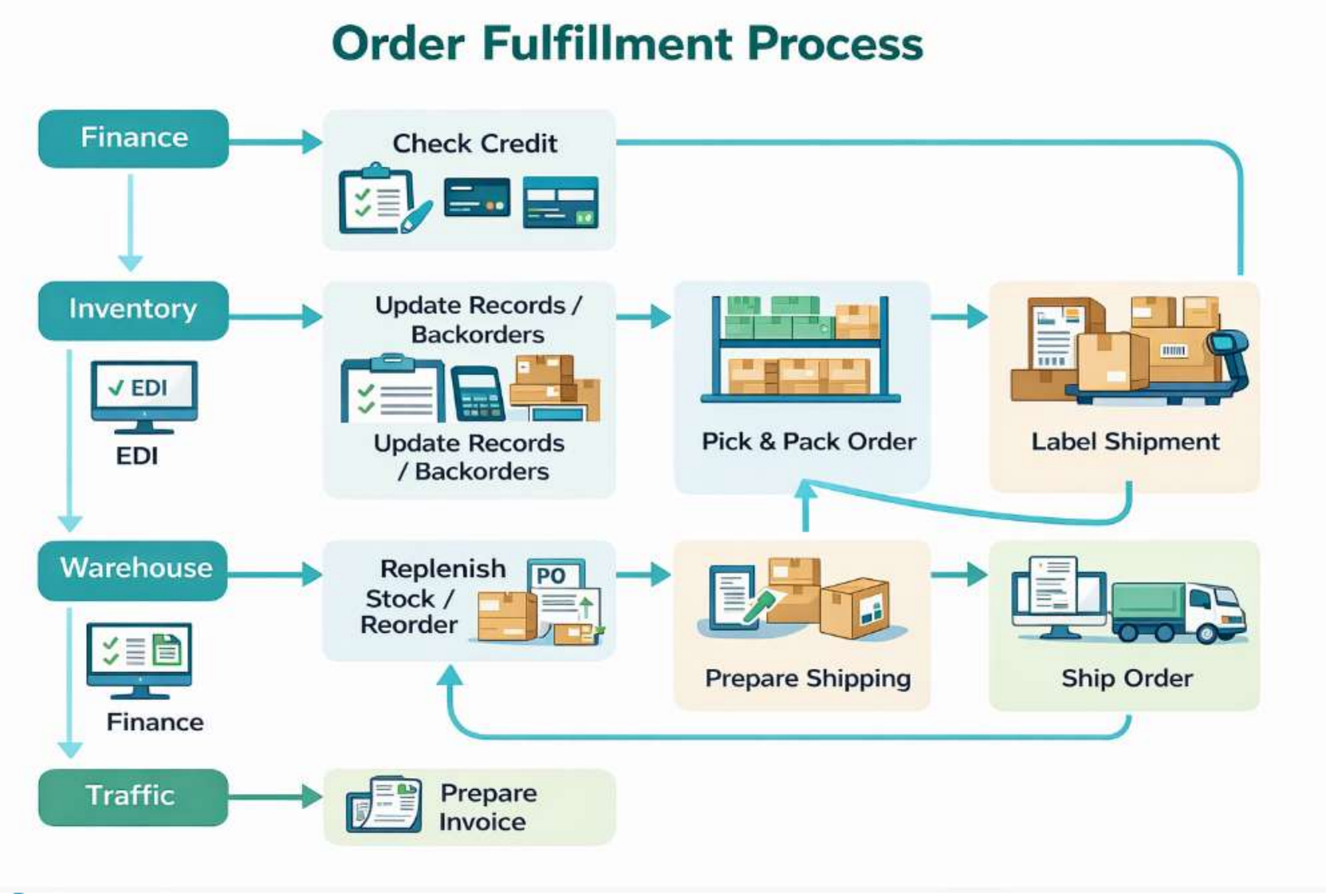
Simple Figure: Flowchart of Order Handling (2. Order Processing) System



Order management

Managerial approaches in order processing

- Order receipt
  - EDI orders bypass checking for completeness as opposite to other order transmissions method
- Order triage refers to classifying orders according to **pre-established guidelines** so that a company can prioritize how orders should be filled
  - Prioritizing attribute(s) delight those that exhibit the chosen attribute and disappoint other customers
- Assigning order to a location
  - Fill an order from closest facility location to customer
  - Facility with largest amount of requested product

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Order Management (6 of 10)

3. Order picking and assembly

1. Order transmittal
2. Order processing
3. Order picking and assembly
4. Order delivery

- Includes all activities from when an appropriate location is authorized to fill the order until goods are loaded aboard an outbound carrier
- Accounts for up to two-thirds of a facility's operating cost and time
- Represents best opportunity to improve the effectiveness and efficiency of an order cycle using improvement:
  - combining multi-order, minimizing movement, positioning frequently ordered items near loading area

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Order Management (8 of 10)

3. Order picking and assembly

- Examples of improvement include:
  - Handheld scanners
  - Radio-frequency identification (RFID) uses electromagnetic tags to identify objects
  - Voice-based order picking
  - Pick-to-light technology
  - Smart glasses with Augmented Reality to help vision picking
  - Robotics and Automation for Automated storage / Automated retrieval (AS/AR)

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Order Management (10 of 10)

4. Order delivery

1. Order transmittal
2. Order processing
3. Order picking and assembly
4. Order delivery

- Order delivery refers to the time from when a transportation carrier picks up the shipment until it is received by the customer (Transportation Chap 12 &13)
- Three key order delivery issues:
  - Variety of options in terms of transit time are now available, such as delivery by 12 noon and 4:30 p.m.
  - Number of shippers are emphasizing both elapsed transit time as well as transit time reliability
  - Transportation carriers are revamping their operations to provide faster and less variable transit times
  - Customers are requiring for delivery windows and measuring delivery reliability

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Customer Service in Logistics (1 of 4)

- Defined as “the ability of logistics management to satisfy users in terms of time, dependability, communication, and convenience.”[2]
- Customer service is much more difficult for competitors to imitate than other marketing mix variables such as price and promotion
- Four dimensions of customer service:
  1. Time
  2. Dependability
  3. Communication
  4. Convenience

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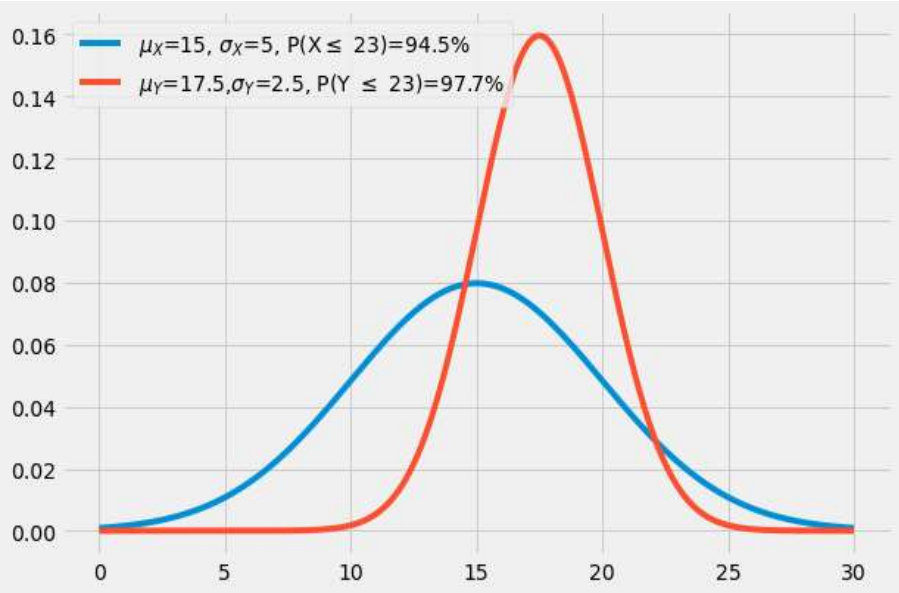
Customer Service (3 of 4)

1. Time
2. Dependability
3. Communication
4. Convenience

- Time refers to period between successive events (e.g., order cycle time) with target to reduce it.
  - Amazon continues to add to the number of locations where it offers *one-hour delivery* for online orders. Example in maintenance as service.
- Dependability refers to 'reliability of encountered service'
  - Consists of three elements:
    - Consistent order cycles (cycle time more regular)
    - Safe delivery (without loss or damage)
    - Complete delivery



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Customer Service (4 of 4)

- Communication
  - To be effective, should be a two-way exchange between seller and customer
  - Goal is to keep both parties informed
  - Requires correct parties to be involved in the process
  - Facilitated nowadays with feedback systems and rating systems
- Convenience
  - Focuses on the ease of doing business with a seller
  - Plays a key role in a consumer's purchasing decision
  - Leads to multichannel marketing and omnichannel retailing

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Managing Customer Service (1 of 2)

1. Goal
2. Objectives
3. Measurement
4. Profitability
5. Failure/Recovery

- Five specific customer service considerations:
  - Establishing customer service goal (specific vs broad)
  - Translate goal into set of SMART objectives:
    - SMART: Specific, Measurable, Achievable, Realistic, Timely
  - Measuring customer service as part of quality-control program
  - Customer profitability analysis (CPA)
  - Service failure and recovery

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Managing Customer Service: setting objectives

1. Goal
2. Objectives
3. Measurement
4. Profitability
5. Failure/Recovery

- Objectives are the means by which goals are to be achieved. Should be “SMART”:
  - Specific, Measurable, Achievable, Realistic, Timely
  - Example: Reduce order picking errors from 5 to 2 percent within a 12-month. Is it SMART?

- Goal and objectives should take into account
  - Customer needs, eg, What services are needed? Which one is most important? How well are we doing? What could be improved?
  - Benchmarking against competitors in performance (fill rate) and process (to get more insight)

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Managing Customer Services

Measuring customer service

1. Goal
2. Objectives
3. Measurement
4. Profitability
5. Failure/Recovery

- You cannot manage what you cannot measure!
- Key issues in measuring customer service involve:
  - Finding data sources (internal and external)
  - Determining factors to measure in relation to time, dependability, communication, and convenience
  - Limiting measurement to few important metrics
  - Communicating with IT to design a dashboard
  - Visualizing the metrics in clear way

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TABLE 7.2 Select Customer Service Measures

Customer Service Dimension	Measure
Time	Order cycle time
	Inquiry response time
Dependability	Perfect order
	On-time delivery
Communication	Customer complaints
	Order status information
Convenience	Returns process
	Response to emergency situations

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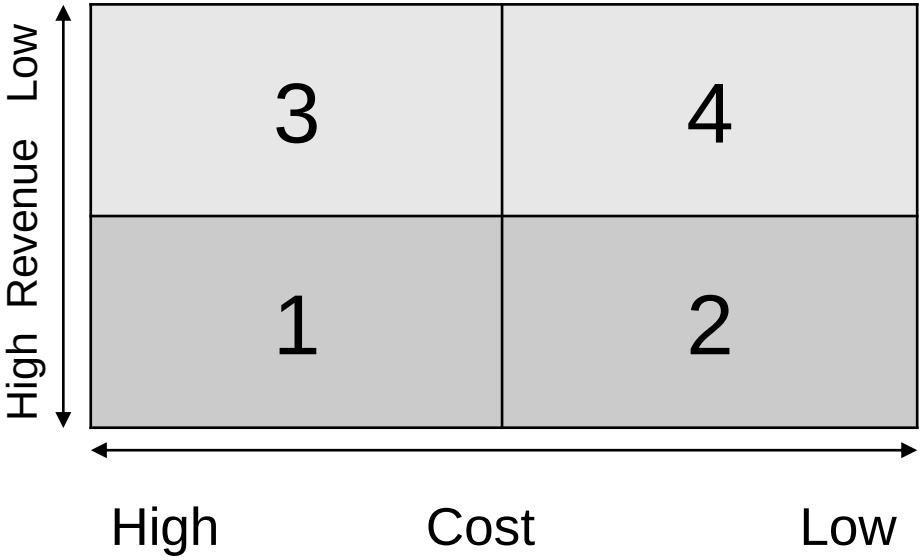
Managing Customer Services

Customer Profitability Analysis (CPA)

1. Goal
2. Objectives
3. Measurement
4. Profitability
5. Failure/Recovery

- **CPA** refers to the allocation of revenues and costs to customer segments or individual customers
- Which one is most important?
- Which one is least important?



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Managing Customer Services

Service Failure and Recovery

1. Goal
2. Objectives
3. Measurement
4. Profitability
5. Failure/Recovery

- 100% service level is hard to realize
- Some situations will occur in which actual performance does not meet the customer's expected performance
- Examples of order-related service failures in logistics include: lost delivery, late delivery, early delivery, damaged delivery, and incorrect delivery quantity
- **Service recovery** refers to a process for returning a customer to a state of satisfaction after a service or product has failed to live up to expectations
- **Service recovery paradox**

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Key Terms (1 of 3)

- Activity-based costing (ABC)
- Benchmarking
- Cause-and-effect forecasting
- Collaborative planning, forecasting, and replenishment (CPFR)
- Customer profitability analysis (CPA)
- Customer service
- Demand management
- Judgmental forecasting
- Make-to-order
- Make-to-stock

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Key Terms (2 of 3)

- Multichannel marketing systems
- Order cycle
- Order delivery
- Order fill rate
- Order management
- Order picking and assembly
- Order processing
- Order to cash cycle
- Order transmittal
- Order triage

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Key Terms (3 of 3)

- Pick-to-light technology
- Service recovery
- Service recovery paradox
- Time series forecasting
- Voice-based order picking

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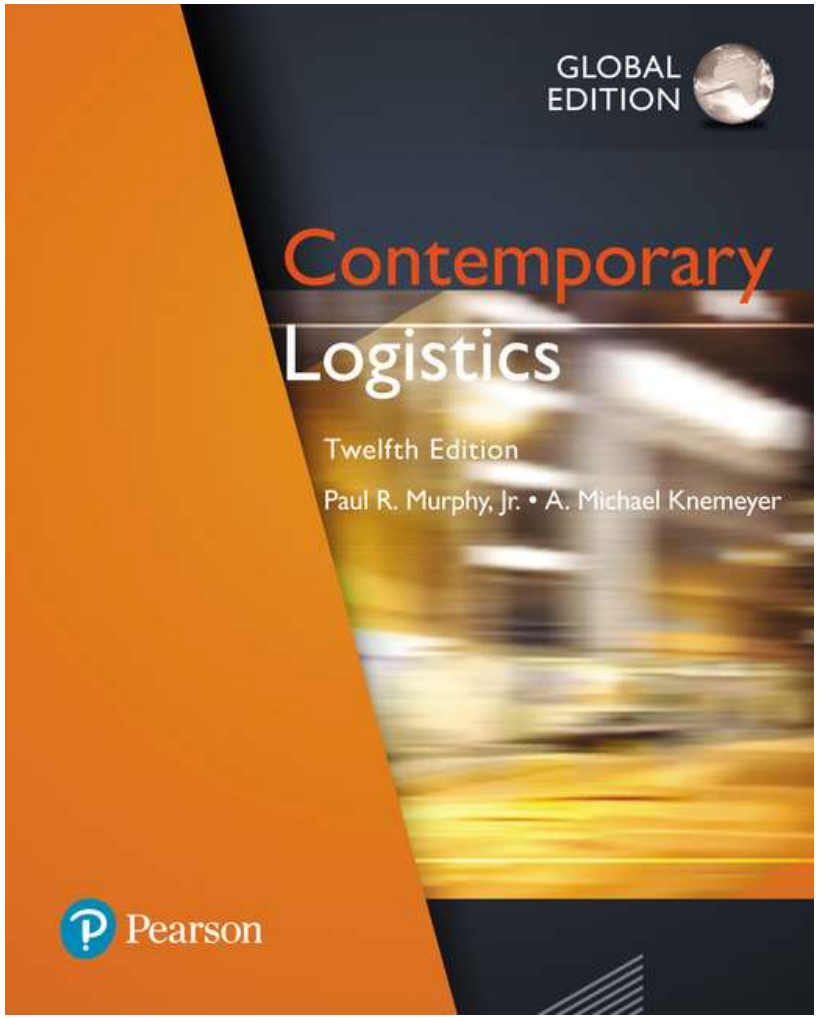
Contemporary Logistics

Twelfth Edition, Global Edition



Chapter 9

Facility Location

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Learning Objectives (1 of 2)

9.1 To examine the strategic importance of facility location

9.2 To explain the general process of determining the optimum number of facilities

9.3 To describe the major factors that influence facility location

9.4 To examine a site's specialized location characteristics

9.5 To explain location decisions using simple grid systems

9.6 To learn about facility relocation and facility closing

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Facility Location

- Facility location
  - Refers to choosing the locations for distribution centers, warehouses, and production facilities to facilitate logistical effectiveness and efficiency
  - Are these strategic or tactical decisions?
- Factors that can influence facility location decisions include:
  - Costs consideration
  - Customer service expectations
  - Location of Customer or Supply Markets

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The Strategic Importance of Facility Location (1 of 3)

- Factors that can influence facility location decisions include:
  - Cost considerations
    - Consumers sensitized to buy only when prices are low
    - Low price/low-cost framework led companies to operate in countries with low-cost and skilled labor
    - In recent years, organizations are reexamining the low-cost labor paradigm (nearsourcing)

Lower production and logistics cost means
more centralization
(few facilities with high utilization)

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The Strategic Importance of Facility Location (2 of 3)

- Factors that can influence facility location decisions include:
  - Customer service expectations
    - Expectations continue to increase over time
    - Customers looking for faster and more reliable order cycles
    - Amazon services orders from more than 120 fulfillment centers located in the United States, Europe, and Asia

Higher customer expectations means
more decentralization
(large network with extra capacity)

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The Strategic Importance of Facility Location (3 of 3)

- Factors that can influence facility location decisions include:
  - Location of customer or supply markets
    - Population shift (Mega Cities)
    - Economic growth (China & India 1/3 of the world)
    - Sustainability (locavore strategy purchasing locally grown or produced foods)

Rank	1970	1990	2014
1	California	California	California
2	New York	New York	Texas
3	Pennsylvania	Texas	Florida
4	Illinois	Pennsylvania	New York
5	Texas	Illinois	Illinois

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Determining the Number of Facilities

- Facilities are generally added when service levels become unacceptable
- Software packages exist to help in determining the number and location of facilities
- Key issue—time it takes to get from facility to majority of the U.S. (or target) population
  - Moving from 1 to 3 facilities saves 1 day lead time
  - Moving from 3 to 10 facilities saves ¼ day lead time[1]

[1]www.chicago-consulting.com

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Figure 9.1: Chicago Consulting's 10 Best Warehouse Networks

THE 10 BEST CHINESE WAREHOUSE NETWORKS
Networks with the Lowest Possible "Time-to-the-Chinese Population"

Number of Warehouses	Average Distance to the Chinese Population (Miles)	Average Transit Lead-Time to the Chinese Population (Days)	Best Warehouse Locations		
ONE	504	3.38	XINYANG		
TWO	377	2.55	LIANYUAN	FEICHENG	
THREE	309	2.15	PINGXIANG	JINAN	ZIYANG
FOUR	265	1.87	PINGXIANG CHANGCHUN	JINING	ZIYANG
FIVE	228	1.65	SHAOGUAN CHANGCHUN	HANDAN NANJING	ZIYANG
SIX	207	1.53	SHAOGUAN CHANGCHUN	HANDAN NANJING	NEIJIANG URUMQI
SEVEN	184	1.42	GUANGZHOU CHANGCHUN HONGHU	HANDAN JINGJIANG	NEIJIANG URUMQI
EIGHT	168	1.31	GUANGZHOU CHANGCHUN HONGHU	LIAOCHENG YIXING BAOJI	YIBIN URUMQI
NINE	154	1.24	BEILIU CHANGCHUN YUEYANG	LIAOCHENG LIYANG BAOJI	YIBIN URUMQI ZHANGZHOU
TEN	141	1.20	BEILIU CHANGCHUN YUEYANG TIANJIN	KAIFENG YIXING BAOJI	YIBIN URUMQI ZHANGZHOU

The 10 Best Chinese Warehouse Networks have been developed based on the lowest possible transit lead-times to "customers" represented by the Chinese population. For example, Xinyang provides the lowest possible lead-time for one warehouse. Any other place will increase transit lead-time to the Chinese population. Similarly putting any three warehouses in any locations other than Pingxiang, Jinan or Ziyang will cause the transit lead-time to be higher than 2.15 days.

© Chicago Consulting
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[illegible]

Chicago Consulting

Source: Courtesy of Terry Harris, Managing Partner of Chicago Consulting.

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General Factors Influencing Facility Location (1 of 7)

- Raw materials, Component parts, Labor, Markets influence on location of manufacturing, processing, or assembly facility
- Location of warehouses, distribution centers, and cross-docking facilities are influenced by locations of plants whose products they handle and markets they serve
- Taxes and incentives, Inventory tax, Incentive packages
- Transportation availability refers to the number of transportation modes (intermodal competition) and the number of carriers within each mode (intramodal competition) and their cost
- Proximity to industry clusters
- Trade patterns and trade agreements
- Quality-of-life considerations (business ecosystem)
- Locating in other countries: Relevant legal system, Political stability, Bureaucratic red tape, Corruption, Protectionism, Nationalism, Privatization, confiscation, Treaties and trade agreements

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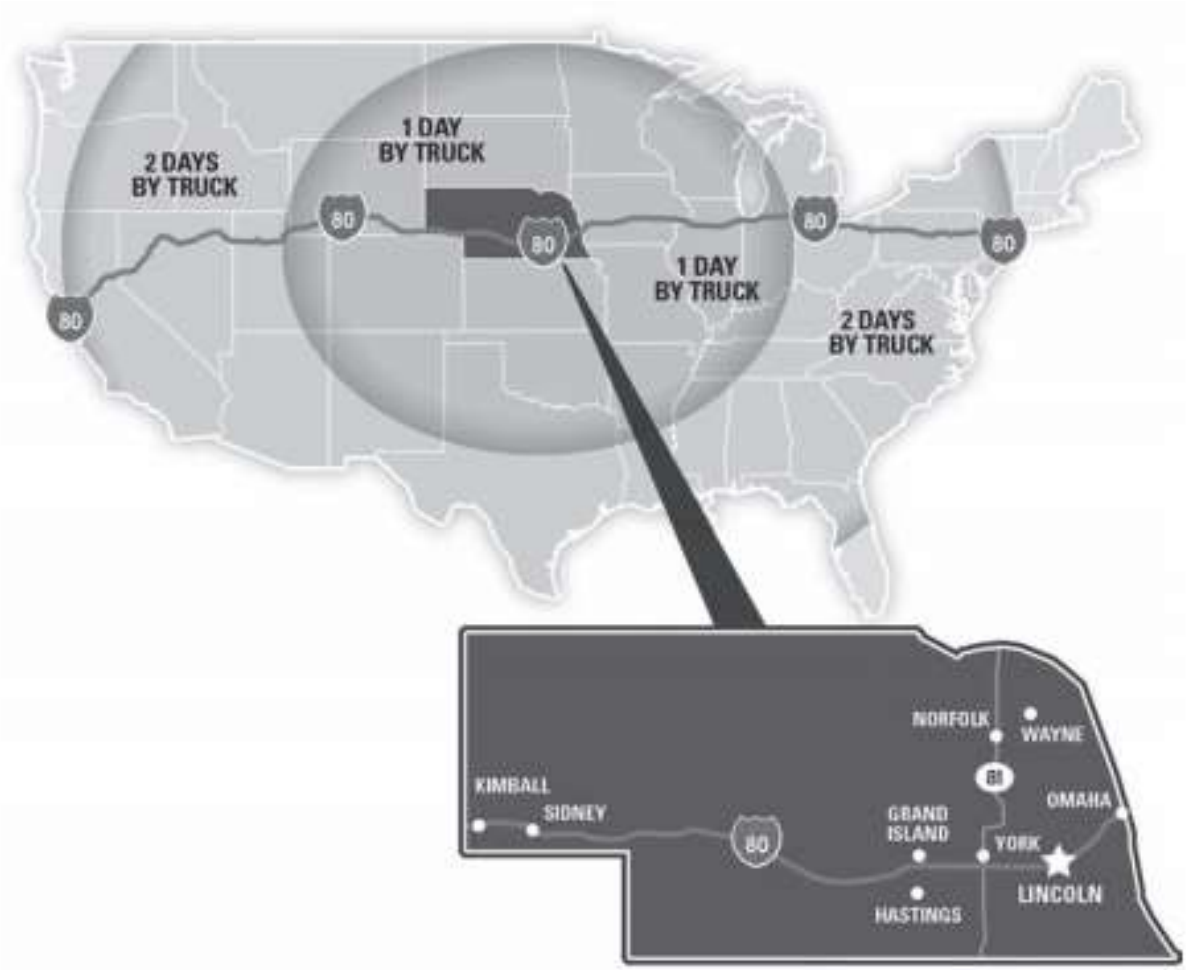
Figure 9.2: Truck Distances from Nebraska



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Specialized Location Characteristics (2 of 4)

- Title search
  - May be needed to make sure a parcel of land can be sold and that there are no liens against it
- Engineers examine site for proper drainage and to determine load-bearing characteristics of the soil
- Due diligence of environmental factors
  - Brownfields
    - Previously used sites that contain chemicals or other types of industrial waste

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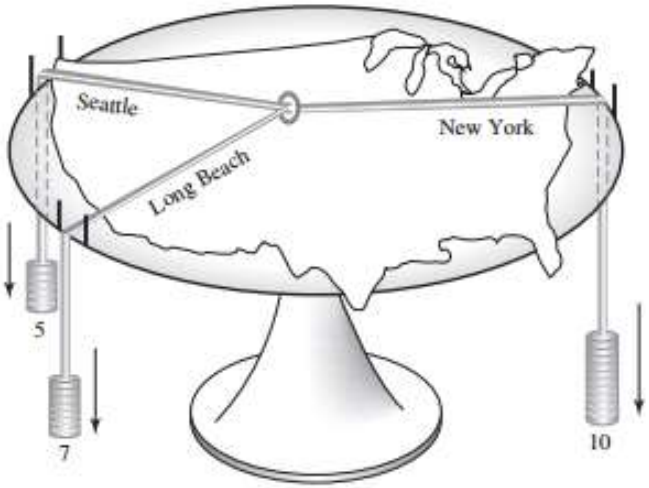
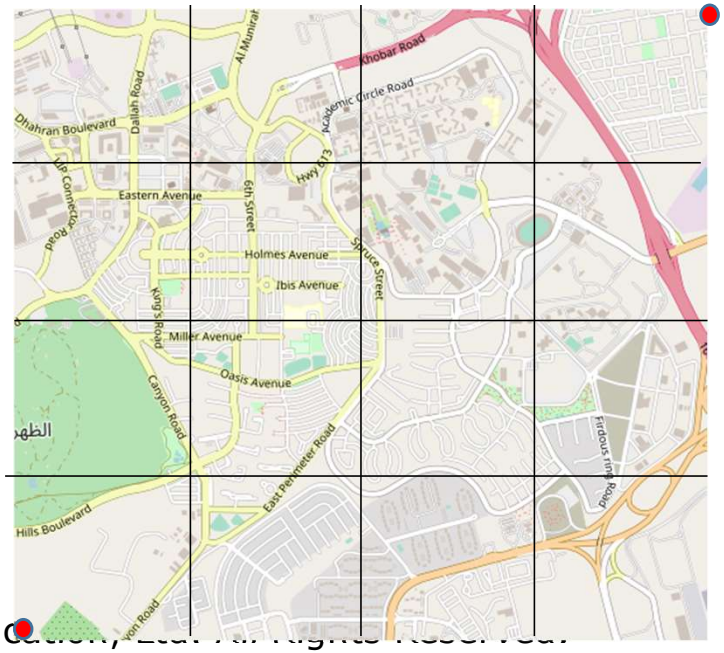
Finding the Lowest-Cost Location Using Grid Systems

- Center-of-Gravity location for a Warehouse (Factory) Serving Retail Stores (Market)
- Grid systems
  - Allow one to analyze spatial relationships with mathematical tools
  - Checkerboard patterns
  - Placed so that they coincide with north–south and east–west line on a map



(50.1594, 26.3181)



(50.1351, 26.2912)

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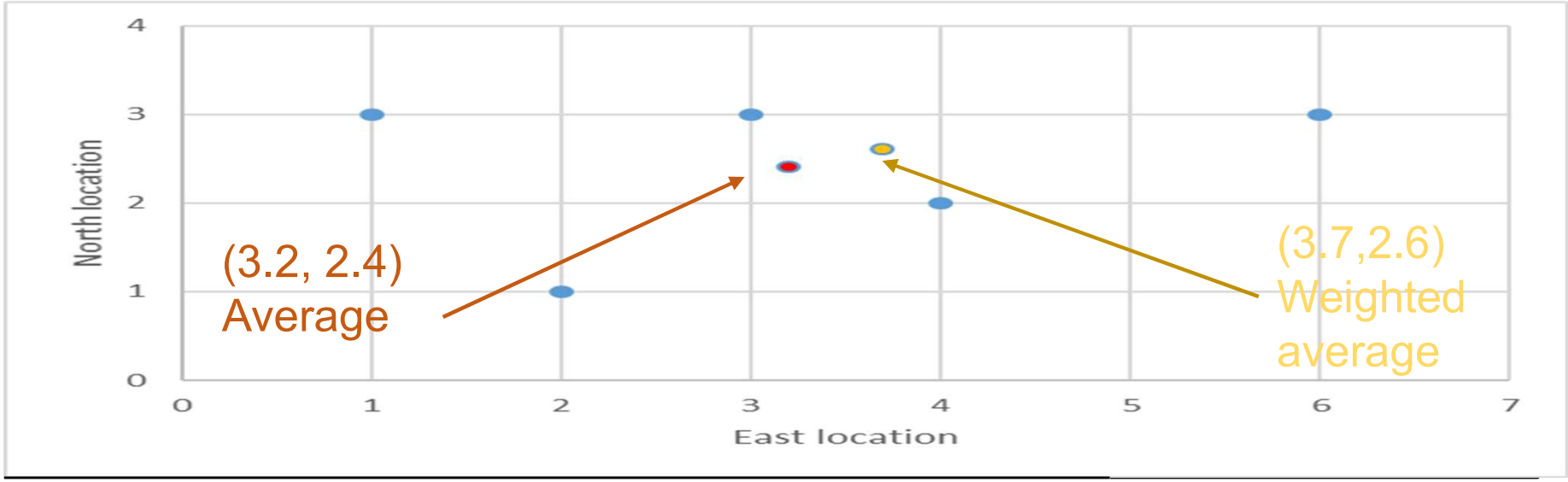
Example: Center-of-Gravity Location for a Warehouse Serving Five Retail Stores



Store	East location	North location	Monthly volume	East x Volume	North x Volume
1	1	3	3	3	9
2	3	3	5	15	15
3	4	2	4	16	8
4	2	1	2	4	2
5	6	3	6	36	18
Total			20	74	52
	3.2	2.4		3.7	2.6
	Average			Weighted average	

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Location Problems

- Single facility location problem:
- Given **m existing facilities**, find the best location of **one new facility**. The best location will be determined based on some distance criteria.
- Solution space
  - Continuous vs Discrete.
- Criteria
  - **Minisum criteria**: the weighted sum of distances among pairs of facilities is minimized.
  - **Minimax criteria**: the weighted maximum distances traveled from new facilities to the existing facilities or to the new facilities is minimized.
  - **Maximin criteria**: the weighted minimum distances traveled from new facilities to the existing facilities or to the new facilities is maximized.

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Location Problems

- Distance measures
  - Rectilinear distance measure :

Distances are measured along paths that are perpendicular to one another. Let (x, y) and (a, b) be two points, then the distance measure is

$d = |x – a| + |y – b|$

  - Euclidean distance measure:

Distances are measured along straight lines. Let (x, y) and (a, b) be two points, then the distance is defined as

$d = [(x - a)^2 + (y - b)^2]^{1/2}$

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Location Problems

- Other distance measures:
  - Squared Euclidean distance measure:
    - $d = [(x - a)^2 + (y - b)^2]$
  - Chebyshev distance measure:
    - $d = \max(|x – a| ; |y – b|)$
  - Actual distance measure:
    - Route distance on Google maps.

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Location Problems

- **Idea**: Given *m* existing facilities, place one new facility
- such that it optimizes a specific criterion.
- **Notations:**
- **$X = (x, y)$** : location of new facility
- **$P_i = (a_i , b_i)$** : location of existing facility *i* , *i* =1, .. ,*m*
- **w_i** : weight associated with travel between new facility and existing facility *i*
- **$d(X, P_i)$** : distance between new facility and existing facility i
- **$f (x, y)$** : the objective function to be minimized

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Location Problems – Rectilinear Minisum

- **Rectilinear Minisum**

$$\text{minimize}: \sum_{i=1}^{m} w_i[|x - a_i| + |y - b_i|]$$

$$\text{subject to}: \; x, \; y \in \mathbb{R}$$

- **The goal is to minimize the total weighted rectilinear distance.**
- **How to solve it?**
  - Median Rule
  - Cumulative weight diagram
  - Resultant force diagram

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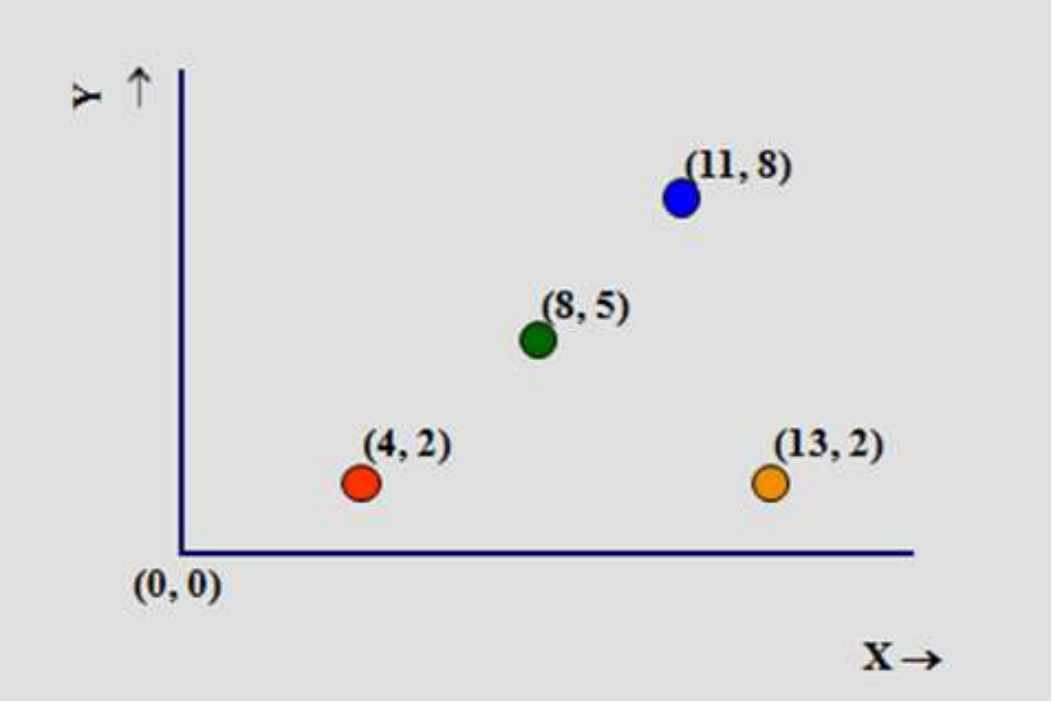
Location Problem: Rectilinear Minisum Example

- Supermarkets 1,2,3, and 4 are located at the points (8,5), (4,2), (11,8), and (13,2) respectively. There are 9, 6, 4, and 12 trips per week between the supermarkets and the new warehouse. Find the best location for the new warehouse based on the rectilinear distance measure and the minisum criteria.



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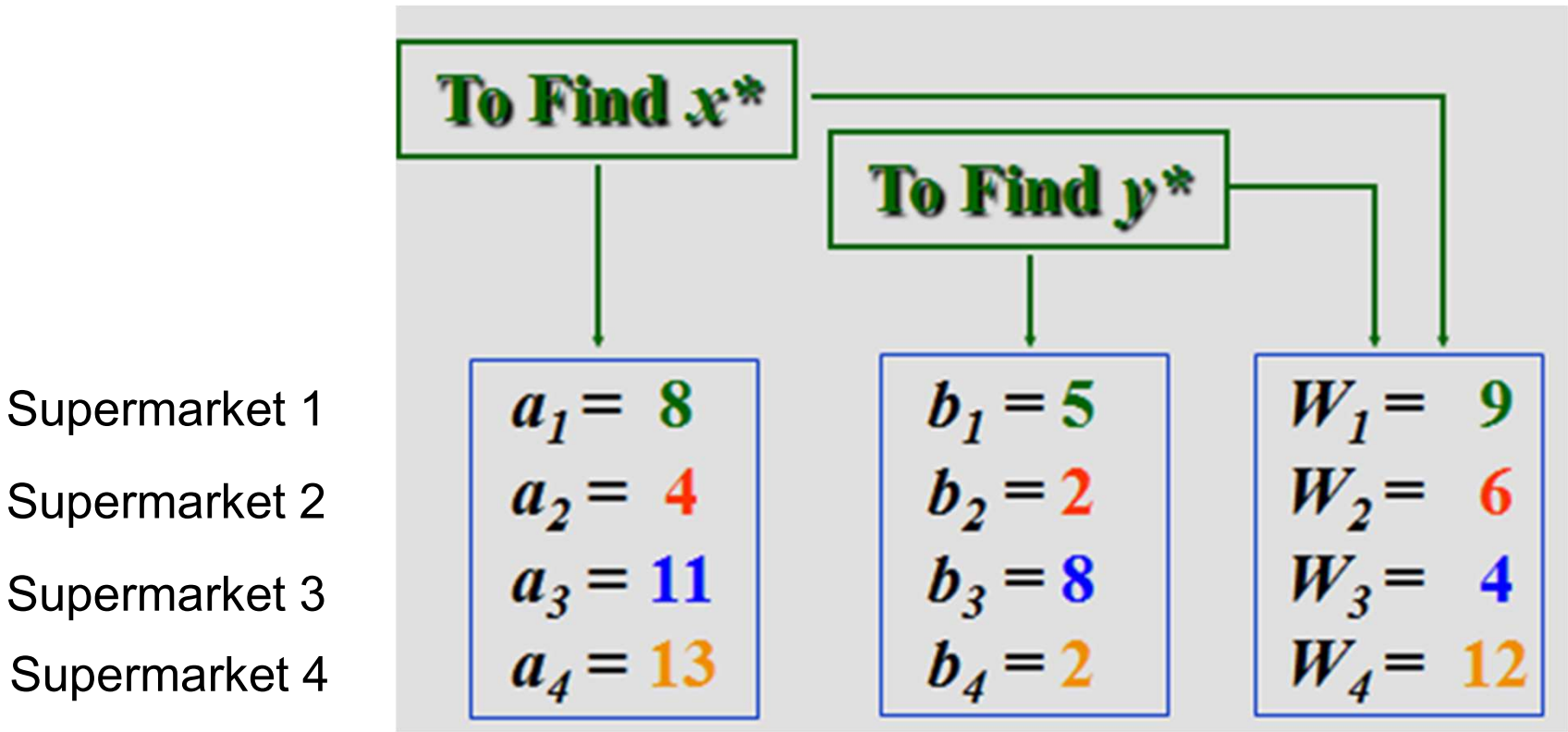
Location Problem: Rectilinear Minisum Example



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Location Problem: Rectilinear Minisum Example

Calculation of Optimal x^*

Supermarket i	Coordinate a_i	Trips w_i	Sum of w_i
2	4	6	6
1	8	9	15
3	11	4	19
4	13	12	31

x^*

$$\frac{1}{2}\sum_{i=1}^{4} W_i = \frac{31}{2} = 15.5$$

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Location Problem: Rectilinear Minisum Example

Calculation of optimal y^*

$$\frac{1}{2}\sum_{i=1}^{4} W_i = \frac{31}{2} = 15.5$$

Supermarket i	Coordinate b_i	Weight w_i	Sum of w_i
2	2	6	6
4	2	12	18
1	5	9	27
3	8	4	31

y^*

Optimal Location: $(x^*, y^*) = (11, 2)$

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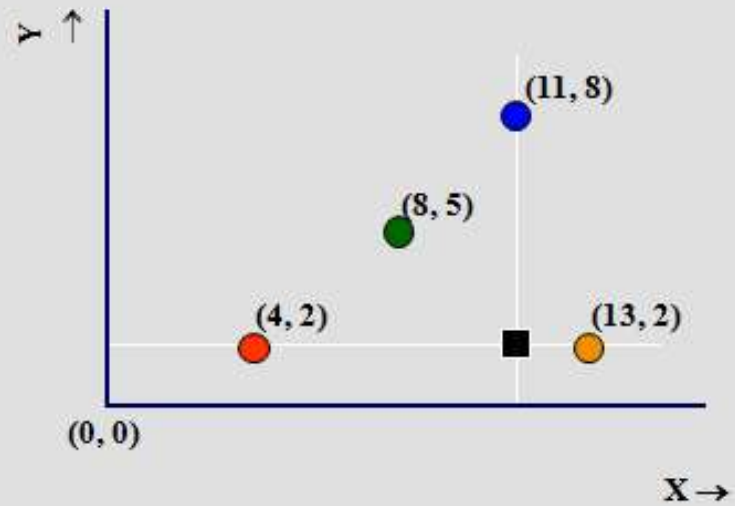
Location Problem: Rectilinear Minisum Example



$$f(x,y) = 9\,|\,11-8\,| + 6\,|\,11-4\,| + 4\,|\,11-11\,| + 12\,|\,11-13\,| +$$
$$9\,|\,2-5\,| + 6\,|\,2-2\,| + 4\,|\,2-8\,| + 12\,|\,2-2\,|$$

$$= 93 + 51$$

$$= 144$$

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Location Problem: Square Euclidean Minisum

- **Square Euclidean Minisum:**

$$\text{minimize}: \sum_{i=1}^{m} w_i[(x-a_i)^2 + (y-b_i)^2]$$

$$\text{subject to}: \; x, \; y \in \mathbb{R}$$

- **The goal is to minimize the total weighted square euclidean distance.**
- **How to solve it?**
  - **Find the min value (set derivative equal to zero and solve)**

$$x^o = \frac{\sum_{i=1}^{m} w_i a_i}{\sum_{i=1}^{m} w_i}$$

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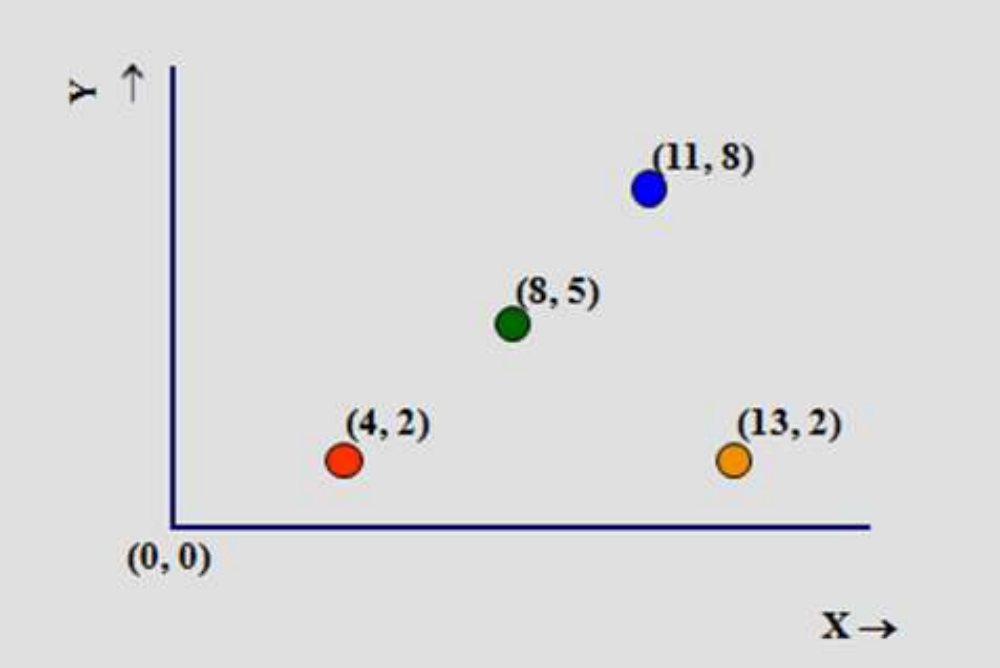
Location Problem: Square Euclidean Minisum Example

- Back to the same example
- Supermarkets 1,2,3, and 4 are located at the points (8,5), (4,2), (11,8), and (13,2) respectively. There are 9, 6, 4, and 12 trips per week between the supermarkets and the new warehouse. Find the best location for the new warehouse based on the square euclidean distance measure and the minisum criteria.



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Location Problem: Square Euclidean Minisum Example

- let $(a_1; b_1) = (8, 5)$, $(a_2, b_2) = (4, 2)$, $(a_3, b_3) = (11, 8)$
- and $(a_4, b_4) = (13, 2)$.
- $w_1 = 9$,$w_2 = 6$,$w_3 = 4$ and $w_4 = 12$.

- $x^* \quad = {\sum_{i=1}^{4} w_i a_i}/{\sum_{i=1}^{4} w_i} = \frac{9*8+6*4+4*11+12*13}{31} = 9.548$
- $y^* \quad = {\sum_{i=1}^{4} w_i b_i}/{\sum_{i=1}^{4} w_i} = \frac{9*5+6*2+4*8+12*2}{31} = 3.645$
- The optimal location w.r.t square euclidean minisum criterion is (x, y) = (9.548, 3.645)

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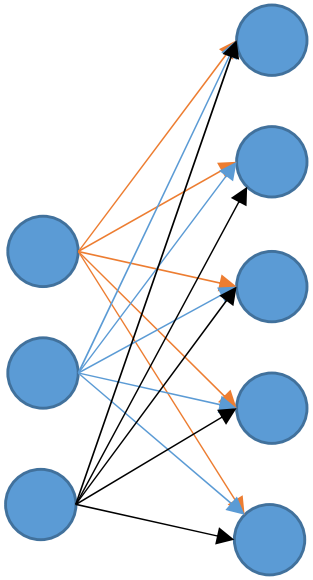
Plants & Markets Allocation Network Optimization



- **Manufacturer of petrochemical with worldwide network in five continents with demand in million liters**
- **Logistics manager has three plants in South America, Asia, and Africa with available capacity (mln liters) to serve demand regions**
- **Cost (inventory, production, transportation) is divided to:**
  1. **Variable cost per mln of liters reported in 1,000$**
  2. **Fixed annual cost (in 1,000$) and capacity**

	Demand Region Production/Transportation Cost ($1,000) per million lt						
Supply Plants	N. America	S. America	Europe	Asia	Africa	Capacity	**Fixed Cost ($)**
S. America	**117**	**77**	**108**	**98**	**100**	20	**6,750**
Asia	**115**	**125**	**90**	**59**	**74**	20	**6,150**
Africa	**142**	**100**	**103**	**105**	**71**	20	**6,000**
Demand (million liters)	12	8	14	16	7		

Mathematical Programming Optimization: Linear Programming (1)

- A furniture dealer deals in only two items–tables and chairs
- Dealer has SR 50,000 to invest and space of at most 60 pieces.
- A table costs SR 2500 and a chair SR 500
- Dealer estimates that the sale of one table makes a profit of SR 250 and of selling one chair SR 75 profit
- Dealer wants to know how many tables & chairs to buy from the available budget to maximize the profit, assuming all items are sold
- Problem formulation
  - Decision variables: x number of tables to buy, y number of chairs to buy
  - Objective function: Maximize $250x + 75y$ (total profit)
  - Constraints: $2500x + 500y \leq 50{,}000$ (Budget constraint)

    $x + y \leq 60$ (storage capacity)
- Problem Type: Linear objective function and constraints in decision variables. Decision variables are continuous. This problem in a linear programming problem
- Solution algorithm: Simplex method implemented numerical software Excel, Lingo, Cplex, Gurobi, Ample, GAMS

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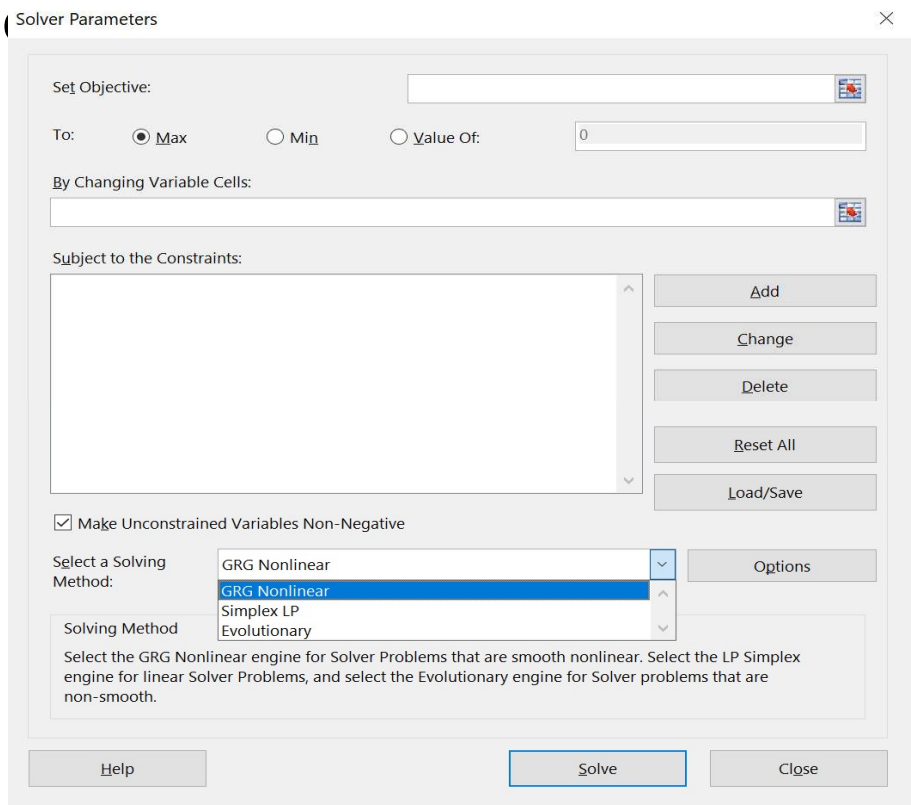
Mathematical Programming Optimization: Linear Programming (2)

- Maximize $250x + 75y$ (total profit)

$$2500x + 500y \leq 50{,}000 \quad \text{(Budget constraint)}$$
$$x + y \leq 60 \text{ (storage capacity)}$$

- Enable Excel Solver Add-in. You find 'Solver' under 'Data'

1. Designate specific Cells for decision variables, objective function, and each of constraints left-hand-sid
2. Configure the 'solver' by setting
3. objective function
4. Selecting types of optimization
5. Setting the decision variables
6. Add the constraint one by one
7. Selecting the algorithm
8. Click Solve



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Plants & Markets Allocation Net. Opt.

- **Decision variables**
  - How much to produce and transport from a plant to a region?
- **Objective**: minimize total costs
- **Constraints**:
  - All demands should be satisfied
  - All plant capacity limits are met

- **Decision variables** x_{ij}
- **Objective**
  Sum $c_{ij}x_{ij}$ for all i, j
- **Constraints**:
  - Unmet demands is 0
  - Left capacity is +

Optimizing the problem using Excel Solver gives

	Demand Region					
Supply Plants	N. America	S. America	Europe	Asia	Africa	Left capacity
S. America	x11 12	x12 8	x13 0	x14 0	x15 0	20 - sum(x11 to x15)>=0
Asia	x21 0	x22 0	x23 4	x24 16	x25 0	20 - sum(x21 to x25)>=0
Africa	x31 0	x32 0	x33 10	x34 0	x35 7	20 - sum(x31 to x35)>=0
Unmet demand	12 - sum(x11 to x31) =0	8 - sum(x12 to x32) =0	14 - sum(x13 to x32) =0	16 - sum(x14 to x34)=0	7 - sum(x15 to x35)=0	

Objective: Sum (c11*x11 to c35*x35) $4,851

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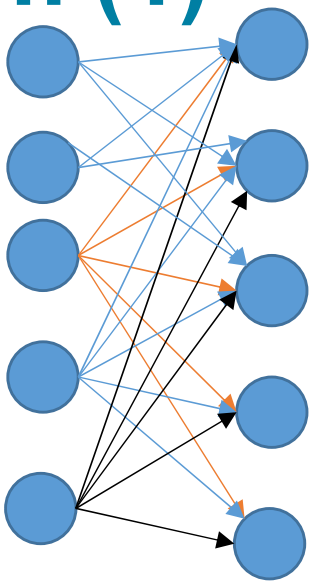
Capacitated Plant Location Problem (1)



- **Manufacturer of petrochemical with worldwide network in five continents with demand in million liters**
- **Logistics manager has <u>five potential</u> plants in North America, South America, Europe, Asia, and Africa with available capacity (mln liters) to serve demand regions**
- **Cost (inventory, production, transportation) is divided to:**
  1. **Variable cost per mln of liters reported in 1,000$**
  2. **Fixed annual cost (in 1,000$) and capacity**

Which plants to open and how to allocate the market demand

	Demand Region Production/Transportation Cost ($1,000) per million lt						
Supply Plants	N. America	S. America	Europe	Asia	Africa	Capacity	**Fixed Cost ($)**
N. America	**81**	**92**	**101**	**130**	**115**	20	**9,000**
S. America	**117**	**77**	**108**	**98**	**100**	20	**6,750**
Europe	**102**	**105**	**95**	**119**	**111**	20	**9,750**
Asia	**115**	**125**	**90**	**59**	**74**	20	**6,150**
Africa	**142**	**100**	**103**	**105**	**71**	20	**6,000**
Demand mln lt	12	8	14	16	7		

Capacitated Plant Location Problem (2)

- **Decision variables**
  - Which plants to open?
  - How much to produce and transport from a plant to a region?
- **Decision variables**
  - z_i: 1 if plant i is open, otherwise.
  - x_{ij}: quantity to ship from i to j
- **Objective**: minimize total costs including variable and fixed costs. <u>Fixed cost is only paid</u> if plant i is open
- **Objective** Sum $f_i z_i$ for all i + Sum $c_{ij} x_{ij}$ for all i, j

  Fixed cost Variable cost
- **Constraints**:
  - All demands should be satisfied
  - All plant capacity limits are met
- **Constraints**:
  - $Unmet\ demand\ is\ 0 \quad (D_j - \sum_{i=1}^{5} x_{ij} = 0)$
  - Left capacity is + $(20z_i - \sum_{j=1}^{5} x_{ij} \geq 0)$

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Capacitated Plant Location Problem (3)

	Demand Region Production/Transportation Cost ($1,000) per million lt						
Supply Plants	N. America	S. America	Europe	Asia	Africa	Left Capacity	**Open / close**
N. America	**X11 12**	**X12 8**	**X13 0**	**X14 0**	**X15 0**	**20z1-sum(x11 to x15)**	**Z1 1**
S. America	**X21 0**	**X22 0**	**X23 0**	**X24 0**	**X25 0**	20z2-sum(x21 to x25)	**Z2 0**
Europe	**X31 0**	**X32 0**	**X33 0**	**X34 0**	**X35 0**	20z3-sum(x31 to x35)	**Z3 0**
Asia	**X41 0**	**X42 0**	**X43 4**	**X44 16**	**X45 0**	20z4-sum(x41 to x45)	**Z4 1**
Africa	**X51 0**	**X52 0**	**X53 10**	**X54 0**	**X55 7**	20z5-sum(x51 to x55)	**Z5 1**
Unmet Demand	12-sum(x11 to x51)	8-sum(x12 to x51)	14-sum(x13 to x53)	16-sum(x14 to x54)	7-sum(x15 to x55)		

Using Excel Solver we find that the optimal cost is $ 23,689

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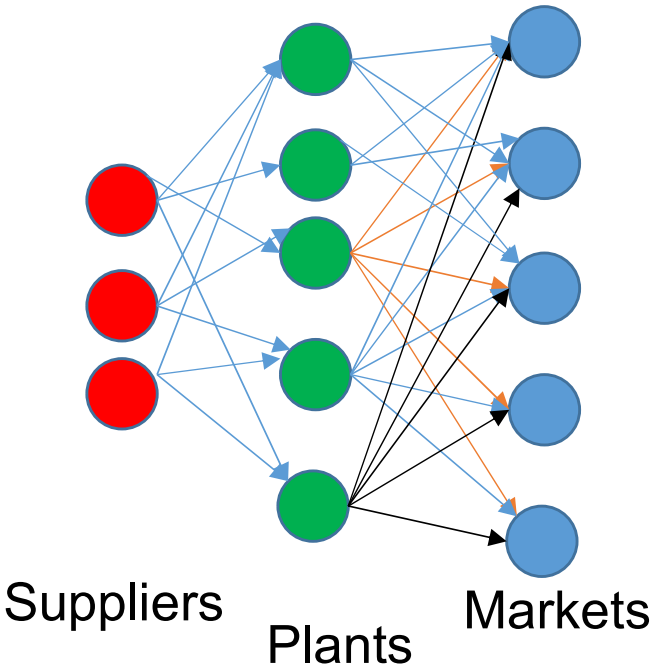
Capacitated Plant Location Problem (4) Two-Echelon Network

- Same decisions as before: which plants to open? how to allocate the markets demand?
- In addition, the logistics manager needs to select the plant suppliers and decide on raw materials quantity bought from each supplier
- Contract with the suppliers is transaction based with variable cost ($1,000 per mln liters) and annual capacity (mln) given as



	Plant Purchasing and Transp. cost per million liters					
Suppliers	N. America	S. America	Europe	Asia	Africa	Capacity
N. America	41	46	51	65	58	40
Asia	59	39	54	49	50	35
Africa	51	53	48	60	56	25

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CPLP: two-echelon

- **Decision variables**
  - Which plants to open?
  - How much to produce and transport from a plant to a region?
  - How much to buy from each supplier?
- **Decision variables**
  - z_i: 1 if plant i is open, otherwise.
  - x_{ij}: quantity to ship from i to j
  - u_{li}: quantity shipped from supplier l to plant i
- **Objective** Sum $f_i z_i$ for all i + Sum $c_{li} u_{li}$ for all l, i+ Sum $c_{ij} x_{ij}$ for all i, j

  Fixed cost Variable cost
- **Constraints**:
  - All demands should be satisfied
  - All plants and suppliers capacity limits are met
  - Output flow from a plant cannot exceed its input flow of raw material
- **Constraints**:
  - $Unmet\ demand\ D_j - \sum_{i=1}^{5} x_{ij} = 0$ and Left plant capacity $20z_i - \sum_{j=1}^{5} x_{ij} \geq 0$
  - Left supplier capacity $C_l - \sum_{i=1}^{5} u_{li} \geq 0$, balancing flows $\sum_{i=1}^{5} u_{li} \geq \sum_{j=1}^{5} x_{ij}$

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CPLP: two-echelon
Excel optimization results

Optimal cost $ 26,340

Plants	N. America	S. America	Europe	Asia	Africa	Left Capacity	Plant Open/Close	Total output Plant
N. America	12	8	0	0	0	0	1	20
S. America	0	0	0	0	0	0	0	0
Europe	0	0	0	0	0	0	0	0
Asia	0	0	4	16	0	0	1	20
Africa	0	0	10	0	7	3	1	17
Unmet Demand	0	0	0	0	0			

	Quantities from suppliers to plants					Excess
Suppliers	N. America	S. America	Europe	Asia	Africa	supplier capacity
N. America	20	0	0	0	0	20
Asia	0	0	0	20	15	0
Africa	0	0	0	0	2	23
Total quantity to plant	20	0	0	20	17	

Possible Extensions of location problems

- Network design with multi-commodity
- Uncertain demand values and uncertain cost parameters
- Including inventory (cycle + safety) cost of end products to satisfy service level constraints
- Taking into account possible disruption in plants and DCs availability
  - In this case deciding optimally on facilities backup to be used in case of unavailable

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Facility Relocation and Facility Closing (2 of 5)

- Facility relocation
  - Occurs when a firm decides that it can no longer continue operations in its present facility and must move operations to another facility to better serve suppliers or customers
  - Common reason for facility relocation
    - Lack of room for expansion at current site

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Facility Relocation and Facility Closing (3 of 5)

- Facility relocation
  - Decision involves a comparison of the advantages and disadvantages of a new site to the advantages and disadvantages of an existing location
  - Must also consider the potential consequences of relocation on their human resources (consequences that may not be easily quantified)
  - Must keep current employees informed
  - Must be prepared for unplanned glitches during transition to prevent customer service issues

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Facility Relocation and Facility Closing (4 of 5)

- Facility closing
  - Occurs when a company decides to discontinue operations at a current site because the operations may no longer be needed or can by absorbed by other facilities
  - Examples leading to facility closure include:
    - Eliminating redundant capacity in mergers and acquisitions
    - Improving supply chain efficiency
    - Poor planning
    - Insufficient volume of business

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Facility Relocation and Facility Closing (5 of 5)

- Facility closing
  - Worker Adjustment and Retraining Notification (WARN) Act
  - Human impact must be considered
    - Employees should be kept informed
    - Individuals are losing jobs and pay
    - Individuals suffer a loss of self-esteem
  - Poorly handled closings can lead to negative publicity

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Key Terms (1 of 2)

- Agglomeration (industry cluster)
- Brownfields
- Center-of-gravity approach
- Expatriate workers
- Facility closing
- Facility location
- Facility relocation
- Free trade zone
- Grid systems
- Intermodal competition
- Intramodal competition
- Inventory tax
- Locavore strategy

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Key Terms (2 of 2)

- Maquiladora plants
- Nearsourcing
- Pure materials
- Quality-of-life considerations
- Right-to-work laws
- Supplier parks
- Sweatshops
- Weight-gaining products
- Weight-losing products

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